Software Quality Engineering – Assignment 2

Manual Functional Testing and Test Reporting

Test Summary Report

Calori – calorie and macro tracker for Android

Version under test: 1.0.0 (build 1), source commit 35e2473 (2 Sep 2026)

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Course	Instructors	Report date
Software Quality Engineering	Mr. Fraz Zaheer, Mr. Talha Dar	11 October 2026

Document Control

Table 1: Document information

Item	Detail
Document	Test Summary Report (with test plan, RTM, test cases, defect log and evidence as appendices)
Application	Calori – Android calorie and macro tracker (open-source Flutter project)
Version under test	1.0.0 (build 1), source commit 35e2473 (2 Sep 2026)
Build	Release APK built from source, SHA-1 f89461552d0e07e4fac8c4471b66fd10bbb60eca
Prepared by	Usman Ghani (24f3000), Faizan Ali (24f3029)
Prepared for	Mr. Fraz Zaheer, Mr. Talha Dar – Software Quality Engineering
Status	Final – for submission

Version history

Table 2: Version history

Version	Date	Author	Change
0.1	11-Oct-2026	Usman Ghani	Test plan, scope and requirement list (FR-001 – FR-046)
0.2	11-Oct-2026	Faizan Ali	Test case design (47 cases), RTM, expected values from the formula oracle
0.3	11-Oct-2026	Both	Execution results for all cases; TC-077 added during execution; 21 defects logged
1.0	11-Oct-2026	Both	Charts, analysis, recommendations and sign-off – final

Abbreviations

Table 3: Abbreviations

Term	Meaning
AUT	Application under test
FR	Functional requirement
TC	Test case
DEF	Defect
RTM	Requirement traceability matrix
EP / BVA / DT / ST / EG	Equivalence partitioning / boundary value analysis / decision table / state transition / error guessing
BMI	Body-mass index (kg ÷ m²)
TDEE	Total daily energy expenditure (maintenance calories)
kcal	Kilocalories
UC-xx	Use case from the project's 04-use-cases.md

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1. Executive Summary

This report presents the results of manual functional testing of **Calori**, an Android calorie and macro tracker, at version 1.0.0 (build 1), the source state of commit 35e2473 dated 2 September 2026. A two-person QA team tested a release build on an Android 16 emulator on 11 October 2026, including the optional photo-estimate service.

We tested eight modules – onboarding, the goal and safety rules, manual food logging, the day view, the calendar, goal management, photo logging and navigation – against 46 functional requirements derived from the project's use cases. All 48 test cases were executed: **32 passed (66.7%) and 16 failed**, none were blocked. We logged **21 defects**: 1 Critical, 12 Major, 4 Minor and 4 Trivial.

What works well. Calorie and macro targets matched the published formula exactly in every scenario we calculated independently; the safety refusals and corrections appear inline with clear wording; food search is fast, works offline and says clearly when nothing matches; the photo path answers 'no food' and 'no connection' honestly and keeps the photo for a retry; a repeated meal gives identical numbers.

Key findings.

- Underweight users cannot finish setting up: both alternatives the app offers them lead to a loading screen that never ends (DEF-06, Critical).
- The under-18 stop – the app's only hard safety block – is bypassed by closing and reopening the app (DEF-02).
- Saved goals are unreliable: switching kg/lb erases the goal date and a hand-set target (DEF-23); the Goal screen sets a target up to 220 kcal away from the rate the user chose (DEF-24); changing the goal rewrites past days' results (DEF-22) and leaves Home's progress ring stale (DEF-21).
- After a food is added by hand, its card stays stuck on 'Saving…' and cannot be used again (DEF-11).
- In the photo flow, proposed portions cannot be corrected before saving (DEF-27), typed details are ignored for gallery photos (DEF-30) and the camera's Close button cannot be tapped (DEF-26).

Quality statement and recommendation: Not ready for release. One Critical defect blocks an entire group of users, and the 9 High-priority defects undermine the two things Calori exists to do – set a safe target and record intake accurately. Release should be reconsidered once the High-priority defects are fixed and the failed test cases pass on re-test (Section 6).

2. Introduction

2.1 Purpose of the report

This report records what was tested in Calori, how it was tested, what was found, and whether the application is functionally fit to release. It is written so that a non-technical reader can follow the summary sections, while the appendices hold the full test plan, traceability, test cases, defect reports and evidence for review.

2.2 Application overview

Calori is an open-source Flutter application for Android that estimates a daily calorie and protein target and lets the user log what they eat. Its main parts are:

- **Onboarding and goal engine** – asks for sex, age, height, weight, target weight and activity, computes a target with the Mifflin-St Jeor equation, and applies safety rules (adults only, no target below BMI 18.5, at most 1 % of bodyweight per week, a calorie floor of 1,500 kcal for men and 1,200 kcal for women).
- **Manual logging** – an offline database of about 21,800 foods from eight national food tables, searchable in English and regional names, with household serving sizes.
- **Day view, calendar and goal screen** – progress against the target per day and per month, editing of logged meals, and changes to the goal with hand-set overrides.

- **Photo logging (optional)** – a meal photo is sent to the developer's own Cloudflare Worker, which asks a vision model for a list of foods; nothing is saved until the user confirms.

2.3 Version and environment under test

Table 4: Version and environment under test

Item	Value
Application / version	Calori 1.0.0 (build 1), source commit 35e2473 (2 Sep 2026)
Build	flutter build apk --release with photo estimates configured; APK SHA-1 f89461552d0e07e4fac8c4471b66fd10bbb60eca
Device	Android Emulator, AVD 'finall' (sdk_gphone64_x86_64), Android 16 / API 36, 1080 × 2424 px at 420 dpi
Photo service	Developer's Cloudflare Worker (health check OK), model google/gemini-2.5-flash via OpenRouter
Network	Online; airplane mode for the offline cases (TC-070)
Test date and time	11-Oct-2026, 20:00–21:23 (local time)

2.4 Modules and features tested

Table 5: Modules and features tested

ID	Module	Features covered	FRs	TCs	Owner
M1	Onboarding – Profile Entry	Sex, age, height and weight entry; unit toggles; under-18 stop; back navigation	6	6	Usman Ghani
M2	Goal Engine & Safety Rules	Target calculation, macro split, BMI refusals, rate and calorie-floor clamps, plan screen	8	10	Usman Ghani
M3	Manual Food Logging	Food search, regional names, no-match handling, portion picker, add to log, recent foods	6	6	Faizan Ali
M4	Day View & Entry Editing	Ring and totals, date strip, entry editor (meal type, portions, remove), delete and undo	6	6	Faizan Ali
M5	Calendar	Month grid, month paging, day detail, month summary, history kept against old targets	5	5	Faizan Ali
M6	Goal Management	Goal steppers, rate, refusals, hand-set calories and protein, unit preferences, Sources	6	6	Faizan Ali
M7	Photo Logging	Viewfinder, gallery photos, details hint, review and correction, save, no-food, offline, repeat meals	7	7	Usman Ghani
M8	Navigation	Nav bar tabs, '+' and calendar buttons, Android Back	2	2	Faizan Ali

Out of scope by definition: all non-functional testing (performance, security, usability, compatibility, accessibility), unit and automated testing, iOS, and the internals of the Worker and of the food-database build tools. The full scope statement is in Appendix A.

3. Test Approach and Strategy

3.1 Overall approach

Testing was requirements-based black-box testing. Requirements were derived from the project's twelve use cases, its README and statements the app makes on screen (Appendix B). Each test case was designed before execution with an expected result taken from the requirement. For calculations, expected values came from an independent re-implementation of the published formula (the 'oracle'), never from the app itself – for example, the reference profile (male, 30, 178 cm, 78 → 72 kg, moderately active) must give 2,160 kcal and 140 g protein.

Every test case was executed by hand, one step at a time, on the emulator. Android's adb tool was used to send taps and text, to read the text shown on screen, to capture screenshots and to record screen video. Scripted test sessions were complemented by six time-boxed exploratory sessions (Appendix E), which found 5 further defects and several usability observations.

3.2 Test design techniques

Table 6: Test design techniques used, with examples

Technique	Test cases using it	Example from this project
Equivalence Partitioning	18	Valid metric and imperial units as separate input classes (TC-006, TC-010); regional versus English food names (TC-027); a meal photo versus a photo with no food (TC-069).
Boundary Value Analysis	9	Age 17 / 18 and 119 (TC-003, TC-004); maintenance band at ±0.5 kg – 78.5 vs 78.6 kg (TC-022); calorie floor 1,499 / 1,500 kcal (TC-058); month paging stops at the current month (TC-048).
Decision Table	9	The UC-02 safety rules as an eight-rule table (Table 7), one or more test cases per rule (TC-004, TC-013–TC-016, TC-018–TC-022, TC-077).
State Transition	15	Onboarding steps and their transitions, including restart and the refusal state (Table 8); entry life-cycle add → edit → delete (TC-033, TC-039, TC-044).
Error Guessing	6	Adding the same food twice (TC-034); restarting after the under-18 refusal (TC-013); taking the alternative offered to an underweight user (TC-077); switching units with a hand-set target (TC-061); losing the connection during photo analysis (TC-070).

Table 7: Decision table for the onboarding safety rules (UC-02)

Condition / action	R1	R2	R3	R4	R5	R6	R7	R8
Age < 18	Y	N	N	N	N	N	N	N
Current BMI < 18.5 and goal is to lose	–	Y	N	N	N	N	N	N
Target BMI < 18.5	–	–	Y	Y	N	N	N	N
Direction of goal	–	lose	lose / keep	gain	lose / gain	lose	lose	any
Rate > 1 % of bodyweight per week	–	–	–	–	Y	–	–	N
Target would fall below calorie floor	–	–	–	–	–	Y	–	N
Maintenance (TDEE) ≤ floor	–	–	–	–	–	N	Y	N
ACTION	Block: refusal screen, no way on	Refuse; offer maintain	Refuse; offer maintain	Refuse; offer healthy-weight gain	Clamp rate; honest date	Clamp to floor; later date	Set to maintain	Accept plan
Test cases	TC-004, TC-013	TC-077	TC-016	TC-018	TC-015, TC-019	TC-020	TC-021	TC-014, TC-022
Result	Pass / Fail	Fail	Pass	Fail	Pass / Fail	Pass	Pass	Pass

Table 8: State transition table for onboarding

From state	Event	Expected next state	Test case	Result
Sex	Choose Male / Female	Age	TC-002	Pass
Age	Valid age ≥ 18, Continue	Height & weight	TC-003	Pass
Age	Age < 18, Continue	Blocked (refusal)	TC-004	Pass
Blocked	App restarted	Blocked	TC-013	Fail
Target	Refused; 'Maintain instead'	Activity → Plan (maintenance)	TC-016	Pass
Target	Underweight; alternative chosen	Activity → Plan	TC-018, TC-077	Fail
Plan	Start tracking	Home (profile saved)	TC-025	Pass

3.3 How test cases were prioritised

Each test case carries a priority (how soon it must run and be fixed if it fails) and a severity (how much harm a failure would do). Priority was set by risk:

- **High** – safety rules, calorie calculations, saving and deleting data, and the core logging paths. A failure here either harms the user or makes the diary wrong.
- **Medium** – input validation, edge cases, secondary displays such as the calendar and Recent foods.

- **Low** – cosmetic or informational behaviour (e.g. the Sources screen).

Because many cases share set-up state (a saved profile, logged meals), execution followed module order. Priority was then used to rank defects and to decide what blocks release.

3.4 Roles and responsibilities

Table 9: Roles and responsibilities

Member	Roll no.	Role	Responsibilities
Usman Ghani	24f3000	Test lead	Test plan and scope; environment set-up (APK build, emulator, test images); design and execution of M1 Onboarding, M2 Goal engine and M7 Photo logging; exploratory sessions EXP-01 and EXP-05; compiling the report.
Faizan Ali	24f3029	Test engineer	Requirement analysis and RTM; design and execution of M3 Manual logging, M4 Day view, M5 Calendar, M6 Goal management and M8 Navigation; exploratory sessions EXP-02, EXP-03, EXP-04, EXP-06; defect-log consolidation and report review.

3.5 Tools used

Table 10: Tools used

Tool	Used for
Flutter SDK + Gradle	Building the release APK from the 35e2473 source
Android Emulator (Android 16)	Running the application under test
Android Debug Bridge (adb)	Taps and text input, reading on-screen text, screenshots (screencap), screen video (screenrecord), airplane mode, force-stop and data reset between scenarios
Python (Pillow, matplotlib, python-docx)	Results log, highlight boxes on screenshots, charts, this document
Formula oracle (oracle.py)	Independent expected values for every calculation test
Wikimedia Commons	Openly licensed meal photos used as photo test data (Appendix A.11)
Microsoft Word / LibreOffice	Report format (.docx) and page-layout checks

4. Test Execution Summary

4.1 Overall metrics

Table 11: Test execution metrics

Metric	Value
Total test cases designed	48 (47 before execution + 1 added during execution: TC-077)
Test cases executed	48
Passed	32
Failed	16
Blocked	0
Not executed	0
Pass percentage	66.7% (32 of 48)
Execution effort (hours)	1.4 h on 11-Oct-2026 (20:00–21:23), including 0.25 h build and set-up
Requirements covered	46 of 46 (100%)
Defects logged	21 (5 from exploratory sessions)
Paid photo-analysis calls used	6 (limit agreed with the project owner: 20)

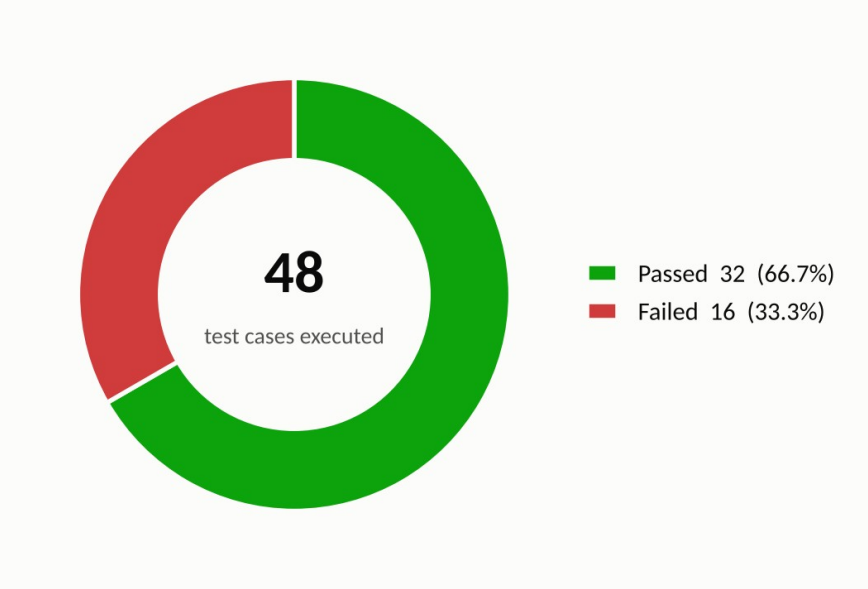


Figure 1: Overall pass/fail result of the 48 executed test cases

4.2 Module-wise breakdown

Table 12: Results by module

Module	Tester	TCs	Pass	Fail	Blocked	Pass %	Defects
M1 Onboarding – Profile Entry	Usman Ghani	6	5	1	0	83%	2
M2 Goal Engine & Safety Rules	Usman Ghani	10	7	3	0	70%	2
M3 Manual Food Logging	Faizan Ali	6	4	2	0	67%	4
M4 Day View & Entry Editing	Faizan Ali	6	5	1	0	83%	3
M5 Calendar	Faizan Ali	5	3	2	0	60%	2
M6 Goal Management	Faizan Ali	6	4	2	0	67%	2
M7 Photo Logging	Usman Ghani	7	3	4	0	43%	5
M8 Navigation	Faizan Ali	2	1	1	0	50%	1
Total		48	32	16	0	67%	21

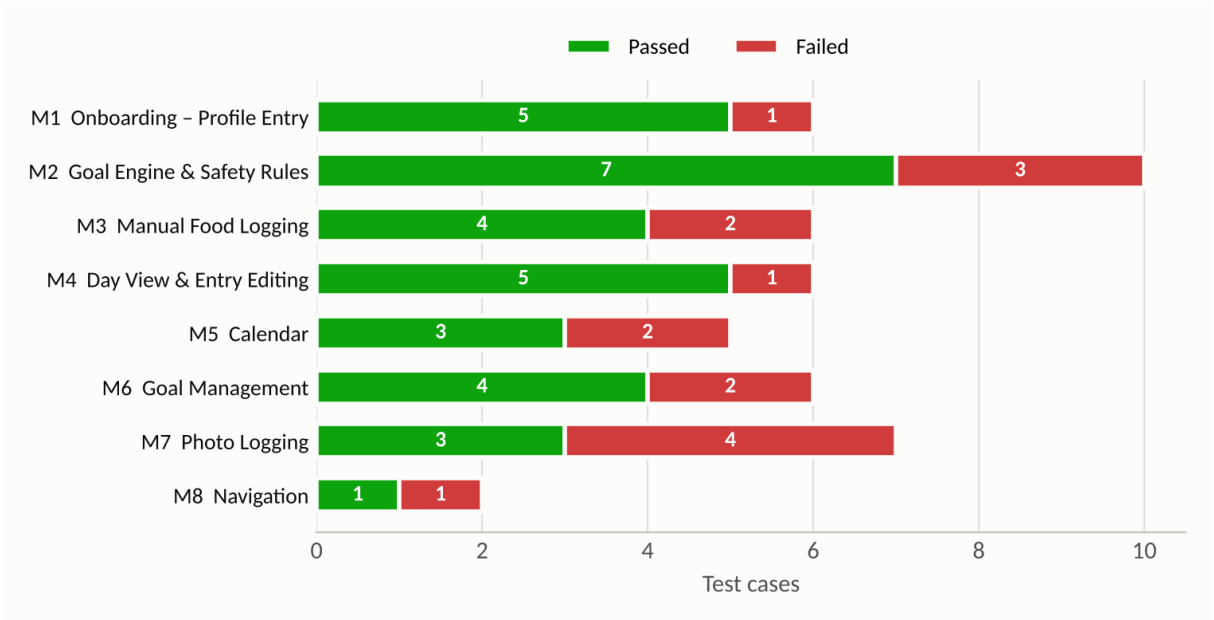


Figure 2: Passed and failed test cases per module

4.3 Requirement coverage

All 46 requirements are covered by at least one test case. 27 requirements are fully met (every linked case passed) and 19 have at least one failing case. The complete traceability matrix is in Appendix B.

4.4 Notes on execution

- TC-077 was added during execution, when exploring the refused loss goal of an underweight user showed that 'Maintain instead' leads to an endless spinner; it is designed, traced and reported like every other case.
- Two test cases had their test data adjusted to match the product: the quantity stepper moves in 0.5 steps, so TC-032 compares 1 with 1.5 servings; TC-066 ran on the review screen of TC-064 so that no extra paid photo call was needed.
- Calculation cases were checked against the oracle; all matched exactly except where a defect is reported (TC-056 / DEF-24).
- Each onboarding scenario started from cleared app data so that earlier answers could not influence the result.

5. Defect Summary

21 defects were logged, each reproduced on the device with screenshot evidence (two also have screen video). All are **Open**: this cycle tested a fixed historical build, so no fixes were delivered during testing.

Table 13: Defects by severity

Severity	Count	Meaning
Critical	1	Blocks a core flow for a group of users with no workaround
Major	12	Wrong result, data loss or a broken feature; a workaround may exist
Minor	4	Incorrect behaviour in validation, wording or a secondary feature
Trivial	4	Cosmetic
Total	21	

Table 14: Defects by priority and by status

Priority	Count	Status	Count
High	9	Open	21
Medium	7	Fixed	0
Low	5	Closed / Rejected / Deferred	0

Table 15: Defects by module and severity

Module	Critical	Major	Minor	Trivial	Total
M1 Onboarding – Profile Entry	0	1	0	1	2
M2 Goal Engine & Safety Rules	1	0	1	0	2
M3 Manual Food Logging	0	2	1	1	4
M4 Day View & Entry Editing	0	1	1	1	3
M5 Calendar	0	1	0	1	2
M6 Goal Management	0	2	0	0	2
M7 Photo Logging	0	4	1	0	5
M8 Navigation	0	1	0	0	1
Total	1	12	4	4	21

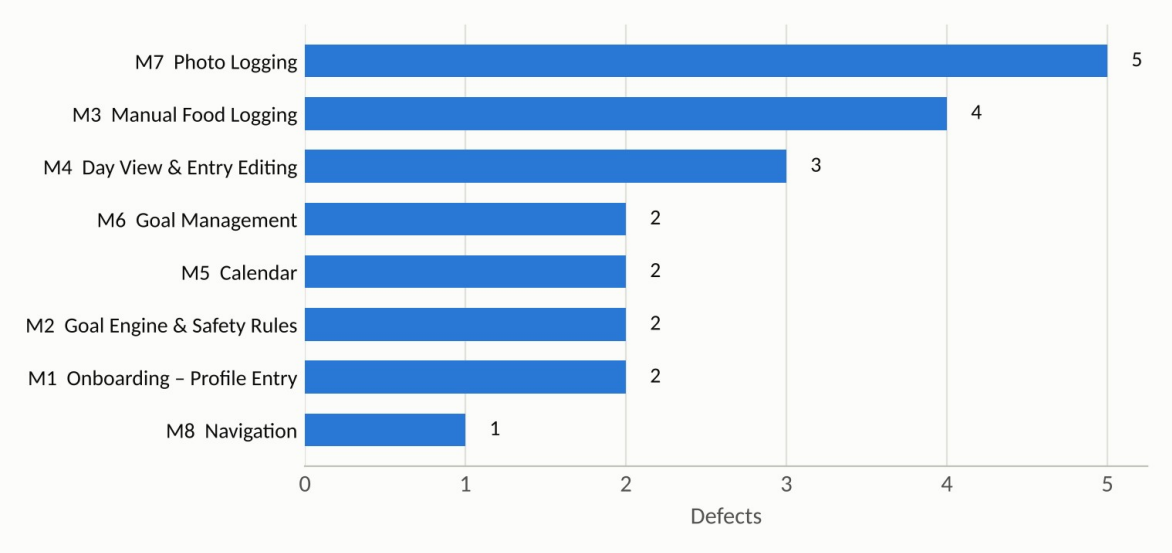


Figure 3: Defects by module

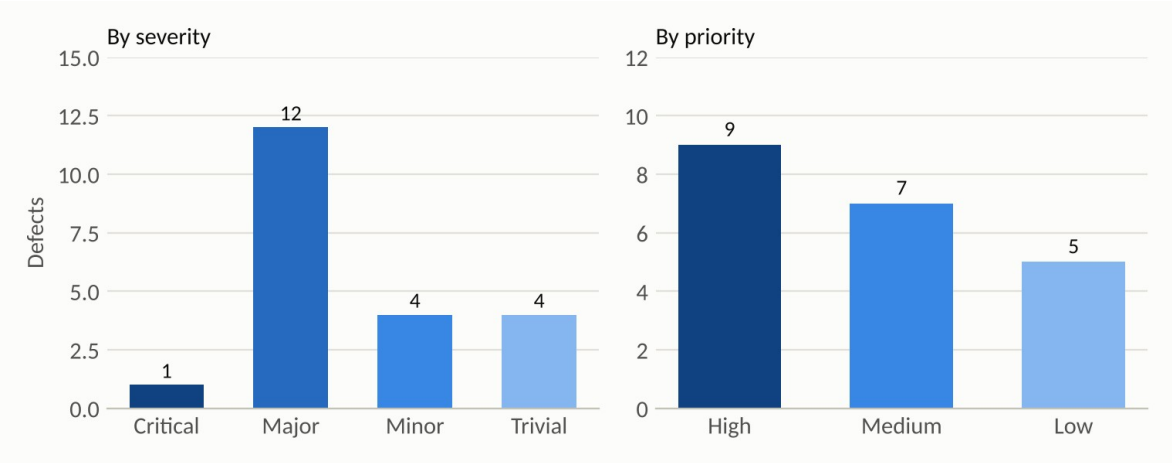


Figure 4: Defects by severity and by priority

5.1 Top five most critical defects

Table 16: Top five most critical defects

ID	Title	Severity / priority	Impact
DEF-06	Underweight users' offered alternatives end on an endless loading spinner	Critical / High	Any user under BMI 18.5 cannot finish onboarding at all, using the very options the app recommends.
DEF-02	Under-18 refusal is forgotten when the app is restarted	Major / High	The app's only hard safety stop can be bypassed in seconds by closing and reopening it.
DEF-23	Switching the weight unit erases the goal date and hand-set calories	Major / High	A display preference silently rewrites the saved plan.
DEF-24	Goal screen ignores the chosen rate (target up to 220 kcal off)	Major / High	The core calculation does not honour the user's input.
DEF-11	'Add to log' stays stuck on 'Saving…' after the first manual add	Major / High	Repeat foods - the most common manual log - cannot be added again without a new search.

Close behind these are DEF-22 (past days re-judged against a new goal), DEF-21 (stale progress ring), DEF-26 (camera cannot be closed) and DEF-27 (photo estimates cannot be corrected before saving). The full defect log and one report per defect are in Appendix D.

6. Observations and Recommendations

6.1 Overall functional quality

Calori's calculation core is sound: every target and macro split we checked against the formula matched to the kilocalorie, and the safety corrections say plainly what changed and why. Offline food search is a strength – fast, regional-name aware, and clear when nothing matches. The weaknesses cluster in three places:

- **Edges of the safety flow** – the underweight alternatives dead-end (DEF-06), the under-18 stop does not survive a restart (DEF-02), system Back closes the app instead of going back (DEF-03), and the 'too fast' correction ignores the goal direction (DEF-07).
- **Consistency of saved data** – a unit preference rewrites the goal (DEF-23), the Goal screen miscalculates from the rate it shows (DEF-24), history is re-judged (DEF-22), displays go stale (DEF-21) and a second manual add gets stuck (DEF-11).
- **The photo review step** – proposed portions cannot be corrected (DEF-27), details are dropped (DEF-30) and the camera screen's own controls are hard to use (DEF-25, DEF-26, DEF-29).

6.2 Recommendations for fixes

- **Fix first (High priority):** DEF-06, DEF-02, DEF-23, DEF-24, DEF-22, DEF-21, DEF-11, DEF-26, DEF-27. Each either harms a user, corrupts the diary or blocks a core action.
- **Then:** the Medium-priority defects, starting with the photo review (DEF-25, DEF-30), data quality of the INDB rows (DEF-13) and system Back handling (DEF-03).
- Show Recent foods for foods logged by hand as well as from photos (DEF-12).
- Store each day's target with the day, so goal changes apply only from today forward (DEF-22) – this also removes the stale-ring class of problem (DEF-21).
- The project's next commit (36ead9d) states that it fixes DEF-11 and DEF-24; both should be re-tested on that build rather than assumed fixed.

6.3 Recommendations for future testing

- **Regression:** re-run all 16 failed cases after fixes, plus the whole M2 and M6 suites, since the goal engine is shared by onboarding and the Goal screen.
- **Real devices:** repeat M7 on physical phones with different status-bar and cut-out heights (DEF-26) and with the real camera rather than the emulator's virtual scene.
- **Usability (out of scope here):** the '+' button covering the last meal's calories and controls hidden behind the floating nav bar were noticed during exploration (Appendix E).
- **Accessibility:** TalkBack labels (several buttons have none), text scaling, contrast of grey text.
- **Performance:** search latency and memory on low-end devices with the 60 MB APK and the 21,800-food database; photo-analysis time on slow networks.
- **Compatibility:** Android 7 (the minimum supported) to 16, tablets and landscape.
- **Security:** the Worker shared secret is compiled into the APK; rate-limit and abuse testing of the Worker.
- **Reliability:** behaviour after process death mid-flow, low storage, and interrupted photo uploads.
- **Automation:** the formula oracle used here could become an automated regression suite for the goal engine.

7. Conclusion and Sign-off

7.1 Completion against exit criteria

Table 17: Exit criteria

Exit criterion (Appendix A.6)	Target	Actual	Met?
Planned test cases executed	100%	100% (48/48)	Yes

Exit criterion (Appendix A.6)	Target	Actual	Met?
Every failure logged as a defect with evidence	100%	16 failures → 21 defects, all with screenshots	Yes
Requirements covered by at least one test	100%	46/46	Yes
Pass rate	≥ 90%	66.7%	No
Open Critical defects	0	1	No
Open High-priority defects	0	9	No

The test cycle is complete: every planned case was run, every requirement was covered and every failure is documented. The quality criteria for release are not met.

7.2 Release recommendation

Not ready for release. Release should be reconsidered when the Critical defect and all High-priority defects are fixed and verified, and the failed test cases pass on a regression run.

7.3 Sign-off

Table 18: Sign-off

Name	Roll no.	Role	Signature	Date
Usman Ghani	24f3000	Test lead		11 October 2026
Faizan Ali	24f3029	Test engineer		11 October 2026

Appendix A – Test Plan

A.1 Objectives

- Verify that every Calori feature behaves as its requirement states.
- Exercise valid, invalid, empty and boundary inputs, error handling, business rules, data handling and navigation.
- Log every deviation as a reproducible defect with evidence, and judge fitness for release.

A.2 Scope

Table 19: Scope

In scope	Out of scope
Onboarding and all UC-02 safety rules Goal calculation and macro split Manual logging and food search Day view, entry editing, delete and undo Calendar Goal management, overrides and units, Sources screen Photo logging through the configured Worker, incl. offline and no-food cases Navigation incl. Android Back	All non-functional testing: performance, security, usability, compatibility, accessibility Unit tests and automated test scripts iOS and other platforms Worker / model internals and prompt quality (only the app's behaviour with the answers is tested) Food-database build tools (tools/build_foods_db) Accuracy of the bundled nutrition data as a whole (clearly implausible rows met during testing are still reported)

A.3 Features and modules to be tested

See Table 5 (Section 2.4) – eight modules, 46 requirements, 48 test cases.

A.4 Test environment

See Table 4 (Section 2.3). One emulator was used throughout; app data was cleared between onboarding scenarios.

A.5 Entry criteria

Table 20: Entry criteria

Criterion	Status on 11-Oct-2026
Release APK of commit 35e2473 builds and installs	Met – built 20:12, installed 20:13
Requirements baselined (FR-001 – FR-046)	Met
Test cases designed and reviewed, expected values computed	Met – 47 cases before execution
Test environment ready (emulator, density 420, photo Worker healthy)	Met – Worker /health returned ok
Test data ready (profiles, food images)	Met – seven images pushed to the emulator gallery

A.6 Exit criteria

- 100% of planned test cases executed.
- Every failure logged as a defect with evidence.
- Every requirement covered by at least one executed test case.
- For a release recommendation: pass rate of at least 90%, no open Critical defects and no open High-priority defects.

A.7 Suspension and resumption

Testing of a module is suspended if the build cannot be installed or a defect blocks every remaining case in it; the photo module would be suspended if the Worker were down or the agreed limit of 20 paid calls were reached. Blocked cases are marked 'Blocked' with the reason. None of these occurred.

A.8 Roles and responsibilities

See Table 9 (Section 3.4).

A.9 Task allocation

Table 21: Task allocation

#	Task	Owner	Start date	End date
1	Test planning, scope, entry/exit criteria	Usman Ghani	11-Oct-2026	11-Oct-2026
2	Requirement analysis and RTM	Faizan Ali	11-Oct-2026	11-Oct-2026
3	Test design – M1, M2, M7 (23 cases)	Usman Ghani	11-Oct-2026	11-Oct-2026
4	Test design – M3, M4, M5, M6, M8 (25 cases)	Faizan Ali	11-Oct-2026	11-Oct-2026
5	Environment set-up: APK build, emulator, test images	Usman Ghani	11-Oct-2026	11-Oct-2026
6	Execution – M1, M2, M7	Usman Ghani	11-Oct-2026	11-Oct-2026
7	Execution – M3, M4, M5, M6, M8	Faizan Ali	11-Oct-2026	11-Oct-2026
8	Exploratory sessions EXP-01 – EXP-06	Both	11-Oct-2026	11-Oct-2026
9	Defect logging, evidence and highlight boxes	Faizan Ali	11-Oct-2026	11-Oct-2026
10	Test summary report, metrics and charts	Usman Ghani	11-Oct-2026	11-Oct-2026
11	Report review and viva preparation	Both	12-Oct-2026	12-Oct-2026

A.10 Risks and mitigations

Table 22: Risks and mitigations

Risk	Mitigation
Photo analysis costs money per call	Agreed limit of 20 calls; gallery images chosen to cover several cases each; 6 used
Emulator differs from a real phone (camera, status bar)	Camera cases use the gallery path; device-specific findings flagged for real-device re-test
Earlier answers leaking between scenarios	App data cleared before every onboarding scenario
Expected values copied from the app's own output	Expected values computed independently by the oracle before execution
Deadline 12-Oct-2026	All execution completed on 11-Oct-2026

A.11 Test data

Profiles: the reference profile (male, 30, 178 cm, 78 kg) and the boundary/decision-table profiles listed in each test case. Photo test data: the following openly licensed images from Wikimedia Commons, pushed to the emulator's gallery.

Table 23: Photo test data and licences

File	Commons title	Licence	Author
biryani.jpg	Hyderabad biryani & naan.jpg	CC0	Abithavelu
egg_breakfast.jpg	Liat Portal for Foodie Disorder - Israeli breakfast plate with fried egg and avocado.jpg	CC BY-SA 4.0	HaJunkiyada
salad.jpg	Garden salad - San Francisco.jpg	CC0	Daderot
dal_rice_chapati.jpg	Bhendi chapati and dal rice.jpg	CC BY-SA 4.0	Wind Hashira
apple.jpg	Red Apple.jpg	CC BY 2.0	Abhijit Tembhekar from Mumbai, India
chapati.jpg	Chapati-1.jpg	CC0	safaritravelplus
cat_nonfood.jpg	Cute gray cat sitting in bed.jpg	CC0	Aleenamustlive

Appendix B – Requirements and Traceability Matrix

Each requirement is traced to the test cases that verify it, their combined result and any defects found.

Table 24: Requirement traceability matrix

FR	Mod.	Requirement	Source	Pri.	Test cases	Result	Defects
FR-001	M1	On first launch with no saved profile the app opens onboarding; Home cannot be reached until a goal has been saved.	UC-01 pre/postcondition	High	TC-025	Pass	—
FR-002	M1	The user selects sex (Male / Female); selecting an option advances to the next step.	UC-01 step 1	High	TC-002	Pass	—
FR-003	M1	Age accepts whole numbers only. An age under 18 ends onboarding on a refusal screen with no path forward.	UC-02 (age < 18)	High	TC-003, TC-004, TC-013	Fail	DEF-02
FR-004	M1	Height must be 100–250 cm and weight 25–400 kg (or their imperial equivalents); out-of-range or malformed values are rejected with a clear message.	UC-01 step 1; error-handling guideline	High	TC-006	Pass	—
FR-005	M1	Height (cm / ft'in) and weight (kg / lb) units are chosen independently; switching a unit converts any value already typed.	README 'Metric or imperial, separately'	Medium	TC-010	Pass	—
FR-006	M1	The user can step back to earlier onboarding questions with answers retained; a progress indicator shows position in the flow.	Onboarding UI	Medium	TC-002	Pass	—
FR-007	M2	The daily target is Mifflin-St Jeor BMR × activity factor, minus (loss) or plus (gain) rate × 7700 kcal/kg ÷ 7, rounded to 10 kcal; protein 1.8 g/kg (to 5 g), fat max(25 % of energy, 0.6 g/kg), carbs the remainder.	UC-01 step 4; README	High	TC-014, TC-056	Fail	DEF-24
FR-008	M2	A target weight below BMI 18.5 is refused inline, stating the lowest healthy weight and offering an alternative: maintenance, or for a gain goal, gaining to a healthy weight.	UC-02 table; engine contract	High	TC-016, TC-018	Fail	DEF-06
FR-009	M2	A loss goal from a user already below BMI 18.5 is refused and maintenance is offered.	UC-02 table	High	TC-077	Fail	DEF-06
FR-010	M2	A rate faster than 1 % of bodyweight per week is clamped to 1 %, with an explanation and the honest achievable date.	UC-02 table	High	TC-015, TC-019	Fail	DEF-07
FR-011	M2	A loss target below the calorie floor (1500 kcal male / 1200 kcal female) is clamped to the floor and the date extended; if maintenance is already at or below the floor, the plan is set to maintain.	UC-02 table	High	TC-020, TC-021	Pass	—
FR-012	M2	A target within ±0.5 kg of current weight is treated as maintenance.	Engine contract	Medium	TC-022	Pass	—
FR-013	M2	Refusal and correction messages use the units the user chose and the correct direction (loss vs gain).	UC-02 'plain language'; README units	Medium	TC-019	Fail	DEF-07
FR-014	M2	The plan screen shows the daily target, macro split, any correction and the projected date; 'Start tracking' saves the profile and opens Home.	UC-01 steps 6–7	High	TC-014, TC-025, TC-077	Fail	DEF-06
FR-015	M3	Search queries the offline food database and shows ranked results, each with a source badge, default serving and kcal; the × button clears the query.	UC-05 steps 3–5	High	TC-028	Pass	—
FR-016	M3	Search matches regional / local names (e.g. channa → chickpea, karela → bitter gourd).	README 'Search that speaks more than English'	Medium	TC-027	Pass	—
FR-017	M3	A query with no matches shows a helpful 'No matches' message; special characters or search operators never cause an error.	UC-05; error handling	Medium	TC-028	Pass	—
FR-018	M3	Before anything is typed, search shows recent and frequent foods; tapping one searches for it.	UC-05 step 2	Medium	TC-030	Fail	DEF-12
FR-019	M3	The portion picker offers serving units first, a quantity of 0.5–20 in 0.5 steps, and shows grams, kcal and macros scaled from per-100 g values.	UC-05 steps 7–8	High	TC-032	Pass	—

FR	Mod.	Requirement	Source	Pri.	Test cases	Result	Defects
FR-020	M3	'Add to log' saves the food (source = database) to the day selected on Home with a meal type defaulted from the time of day, confirms it, returns to Home, and further foods can then be added.	UC-05 step 9	High	TC-033, TC-034	Fail	DEF-11
FR-021	M4	Home shows the selected day's calories consumed against target (ring % and 'X / Y kcal'), protein against target, carbs and fat, and the day's entries.	UC-07	High	TC-033, TC-036	Pass	—
FR-022	M4	Being over target is shown neutrally ('over today'); no negative numbers.	UC-07 'Over target renders neutrally'	Medium	TC-037	Pass	—
FR-023	M4	The date strip shows the last 12 days ending today (no future days) and switches the day shown.	Home UI; UC-05 retroactive logging	Medium	TC-038	Pass	—
FR-024	M4	Tapping an entry opens an editor to change its meal type, rescale item portions and remove items; totals update immediately.	UC-09	High	TC-039	Pass	—
FR-025	M4	Swiping an entry deletes it with an Undo that restores it; 'Remove this meal' deletes it.	UC-09	High	TC-044	Pass	—
FR-026	M4	Home figures update immediately when the goal changes.	UC-07 / UC-10	High	TC-045	Fail	DEF-21
FR-027	M5	The calendar shows the visible month: logged days as progress rings, unlogged past days as a neutral dot, future days dimmed and not selectable.	UC-08 steps 1–3	High	TC-046	Pass	—
FR-028	M5	The user can page to previous months; forward paging stops at the current month.	UC-08 step 5	Medium	TC-048	Pass	—
FR-029	M5	Tapping a day shows its detail (consumed vs target, macros, entries); tapping it again closes the detail.	UC-08 step 4	Medium	TC-049	Fail	DEF-17
FR-030	M5	The month summary shows average kcal per logged day and the number of logged days.	Calendar UI	Medium	TC-051	Pass	—
FR-031	M5	Past days keep the target that was in force when they were logged; goal changes apply from today forward.	UC-10; Goal screen text	High	TC-052	Fail	DEF-22
FR-032	M6	The Goal screen shows the saved target, macros and a caption, and lets the user change current weight, goal weight and rate (0.1–1.0 % per week) with live recalculation and automatic saving.	UC-10	High	TC-053	Pass	—
FR-033	M6	The recalculated target honours the rate chosen on the rate stepper.	UC-10	High	TC-056	Fail	DEF-24
FR-034	M6	Safety rules reapply on the Goal screen, including refusals; a refused goal is not saved.	UC-10	High	TC-057	Pass	—
FR-035	M6	Calories and protein can be set by hand; the engine's suggestion and a Reset stay visible; a value below the floor shows an advisory; unusable values are rejected with feedback; the macro split stays consistent with the calorie target.	README 'A daily target you can overrule'	Medium	TC-058	Pass	—
FR-036	M6	Weight and height units can each be changed on the Goal screen without altering the goal itself.	Goal screen 'About' rows	Medium	TC-061	Fail	DEF-23
FR-037	M6	'Where the food data comes from' opens the Sources screen listing each data source with its attribution.	README data sources	Low	TC-062	Pass	—
FR-038	M7	The camera button opens a viewfinder with a shutter, a Gallery button and an optional 'Add details' field that never takes focus automatically.	UC-03 / UC-04 constraint	High	TC-063	Fail	DEF-25, DEF-26
FR-039	M7	A meal photo is analysed into proposed items (kcal, grams, macros); nothing is logged until Save; Save logs the meal (source = AI) with its photo.	UC-03	High	TC-064	Fail	DEF-29
FR-040	M7	Details typed before capture are sent with the photo and applied to the portions.	UC-04	Medium	TC-065	Fail	DEF-30
FR-041	M7	Before saving, the user can correct proposed items: adjust the portion, edit name or macros, remove an item, and add an item the photo missed.	UC-06	High	TC-066	Fail	DEF-27
FR-042	M7	A photo with no food returns 'No food in that photo' with Try again and manual options, never an invented entry.	README; UC-03	Medium	TC-069	Pass	—
FR-043	M7	Analysis failures (offline, server error, timeout) show a specific message, Try again (photo retained) and 'Add by hand instead'.	UC-11	High	TC-070	Pass	—

FR	Mod.	Requirement	Source	Pri.	Test cases	Result	Defects
FR-044	M7	A repeat meal reuses saved values so the same food gives the same numbers; accepted photo foods feed the Recent suggestions.	UC-12; UC-05 step 2	Medium	TC-072	Pass	—
FR-045	M8	The nav bar switches between Home and Goal with the active tab highlighted; '+' opens Add food; the calendar button opens the calendar.	Shell UI	High	TC-046, TC-074	Pass	—
FR-046	M8	The Android system Back button navigates back within the app (previous onboarding step, or Home from a secondary screen) instead of leaving it.	Android navigation convention	Medium	TC-076	Fail	DEF-03

Appendix C – Test Cases

C.1 Test case summary

Table 25: Test case summary

ID	Module	Req.	Title	Technique	Type	Pri.	Sev.	Tester	Status	Defects
TC-002	M1	FR-002, FR-006	Selecting sex advances to the age step	EP	Positive	High	Major	Usman	Pass	—
TC-003	M1	FR-003	Valid age boundaries 18 and 119 are accepted	BVA	Positive	High	Major	Usman	Pass	—
TC-004	M1	FR-003	Age 17 ends onboarding on the refusal screen	BVA	Negative	High	Critical	Usman	Pass	—
TC-006	M1	FR-004	Valid metric height and weight proceed	EP	Positive	High	Major	Usman	Pass	—
TC-010	M1	FR-005	Unit toggles convert typed values, independently	EP	Positive	Medium	Minor	Usman	Pass	—
TC-013	M1	FR-003	Under-18 refusal persists after the app is restarted	EG / ST	Negative	High	Major	Usman	Fail	DEF-02
TC-014	M2	FR-007, FR-014	Reference profile produces the published target and macros	EP (business rule)	Positive	High	Major	Usman	Pass	—
TC-015	M2	FR-010	Too-fast loss rate is clamped to 1 %/week	DT	Negative	High	Major	Usman	Pass	—
TC-016	M2	FR-008	Target below BMI 18.5 is refused; 'Maintain instead' works	DT	Negative	High	Critical	Usman	Pass	—
TC-018	M2	FR-008	Underweight user's gain to a still-underweight target is offered a healthy-weight gain	DT	Negative	High	Major	Usman	Fail	DEF-06
TC-019	M2	FR-010, FR-013	Too-fast GAIN rate is clamped with gain wording	DT	Negative	Medium	Minor	Usman	Fail	DEF-07
TC-020	M2	FR-011	Loss target below the female floor is clamped to 1,200 kcal	DT / BVA	Negative	High	Major	Usman	Pass	—
TC-021	M2	FR-011	Maintenance already at or below the floor sets the plan to maintain	DT (edge)	Edge	Medium	Major	Usman	Pass	—
TC-022	M2	FR-012	Maintenance band boundary ±0.5 kg	BVA	Edge	Medium	Minor	Usman	Pass	—
TC-025	M2	FR-014, FR-001	'Start tracking' saves the profile and opens Home (survives restart)	ST	Positive	High	Critical	Usman	Pass	—
TC-027	M3	FR-016	Regional names find English foods (channa, karela)	EP	Positive	Medium	Minor	Faizan	Pass	—
TC-028	M3	FR-017, FR-015	Unknown query shows 'No matches'; × clears it	EP	Negative	Medium	Minor	Faizan	Pass	—
TC-030	M3	FR-018	Recent foods are shown before typing after foods were logged by hand	ST	Positive	Medium	Minor	Faizan	Fail	DEF-12
TC-032	M3	FR-019	kcal, grams and macros scale with quantity and serving	EP (business rule)	Positive	High	Major	Faizan	Pass	—
TC-033	M3	FR-020, FR-021	'Add to log' saves to Home with a time-based meal type	ST	Positive	High	Critical	Faizan	Pass	—
TC-034	M3	FR-020	A second food can be added after returning to Add food	ST / EG	Positive	High	Major	Faizan	Fail	DEF-11
TC-036	M4	FR-021	Ring, calorie line and macros reflect logged food	EP (business rule)	Positive	High	Major	Faizan	Pass	—
TC-037	M4	FR-022	Over target shown neutrally, never negative	BVA	Edge	Medium	Minor	Faizan	Pass	—
TC-038	M4	FR-023	Date strip shows 12 days ending today, no future days	BVA	Positive	Medium	Minor	Faizan	Pass	—
TC-039	M4	FR-024	Changing an entry's meal type in the editor	ST	Positive	High	Major	Faizan	Pass	—

ID	Module	Req.	Title	Technique	Type	Pri.	Sev.	Tester	Status	Defects
TC-044	M4	FR-025	'Remove this meal' deletes the entry and closes the editor	ST	Positive	Medium	Major	Faizan	Pass	—
TC-045	M4	FR-026	Home updates when the calorie target is changed	ST	Positive	High	Major	Faizan	Fail	DEF-21
TC-046	M5	FR-027, FR-045	Calendar opens with rings for logged days and dots for unlogged days	EP	Positive	High	Major	Faizan	Pass	—
TC-048	M5	FR-028	Paging months: back works, forward stops at the current month	BVA / ST	Positive	Medium	Minor	Faizan	Pass	—
TC-049	M5	FR-029	Day detail shows consumed vs target, macros and entries	ST	Positive	Medium	Minor	Faizan	Fail	DEF-17
TC-051	M5	FR-030	Month summary averages over logged days only	EP (business rule)	Positive	Medium	Minor	Faizan	Pass	—
TC-052	M5	FR-031	A past day keeps the target it was logged against	DT (business rule)	Negative	High	Major	Faizan	Fail	DEF-22
TC-053	M6	FR-032	Goal screen shows the saved plan	EP	Positive	High	Major	Faizan	Pass	—
TC-056	M6	FR-033, FR-007	Target honours the chosen rate (95 → 92 kg at 1.0 %/week)	EP / BVA (business rule)	Positive	High	Major	Faizan	Fail	DEF-24
TC-057	M6	FR-034	Unsafe goal weight is refused on the Goal screen and not saved	DT	Negative	High	Major	Faizan	Pass	—
TC-058	M6	FR-035	Hand-set calories: suggestion, floor advisory boundary and Reset	ST / BVA	Positive	Medium	Minor	Faizan	Pass	—
TC-061	M6	FR-036	Changing the weight unit keeps the goal and any hand-set values	EG / ST	Negative	High	Major	Faizan	Fail	DEF-23
TC-062	M6	FR-037	Sources screen lists data sources and returns	EP	Positive	Low	Minor	Faizan	Pass	—
TC-063	M7	FR-038	Viewfinder opens; details field not focused; close returns	EP	Positive	High	Major	Usman	Fail	DEF-26, DEF-25
TC-064	M7	FR-039	Gallery meal photo → proposed items → Save logs an AI entry with photo	ST	Positive	High	Critical	Usman	Fail	DEF-29
TC-065	M7	FR-040	Details typed before choosing a gallery photo are applied	EP	Positive	Medium	Minor	Usman	Fail	DEF-30
TC-066	M7	FR-041	Portion of a proposed item can be adjusted before saving	EP	Positive	High	Major	Usman	Fail	DEF-27
TC-069	M7	FR-042	A photo with no food is reported as such	EP	Negative	High	Major	Usman	Pass	—
TC-070	M7	FR-043	Offline failure shows a specific message; retry keeps the photo	EG / ST	Negative	High	Major	Usman	Pass	—
TC-072	M7	FR-044	The same meal photographed again gives the same numbers	EP	Positive	Medium	Minor	Usman	Pass	—
TC-074	M8	FR-045	Nav bar switches Home and Goal with the active tab highlighted	ST	Positive	High	Major	Faizan	Pass	—
TC-076	M8	FR-046	System Back from Calendar / Add food / Goal returns to Home	EG	Negative	Medium	Minor	Faizan	Fail	DEF-03
TC-077	M2	FR-009, FR-014	'Maintain instead' for an underweight user produces a maintenance plan	DT / EG	Negative	High	Critical	Usman	Fail	DEF-06

C.2 Detailed test cases

Every case lists the fields the assignment requires. All cases were executed on 11 October 2026.

Test Case ID	TC-002
Module / Feature	M1 Onboarding – Profile Entry
Linked requirement(s)	FR-002, FR-006
Title / Description	Selecting sex advances to the age step
Technique / Type	EP · Positive
Preconditions	Onboarding open on the sex step.
Test steps	1. Tap 'Male'.
Test data	Sex = Male
Expected result	The age question 'How old are you?' opens, the progress bar advances and a back arrow appears.
Actual result	Tapping Male opened 'How old are you?' with the numeric keyboard up; progress bar advanced from 20% to 40% and a back arrow appeared.
Status	Pass
Priority / Severity	High / Major
Tester / Execution date	Usman Ghani · 11-Oct-2026
Evidence	TC-002_age_step
Defects	—

Test Case ID	TC-003
Module / Feature	M1 Onboarding – Profile Entry
Linked requirement(s)	FR-003
Title / Description	Valid age boundaries 18 and 119 are accepted
Technique / Type	BVA · Positive
Preconditions	Onboarding on the age step.
Test steps	1. Type 119 and observe Continue. 2. Clear the field (relaunch after pm clear), type 18 and tap Continue.
Test data	Age = 119; Age = 18
Expected result	For both values Continue is enabled; with 18, Continue opens 'Your height and weight'.
Actual result	119 enabled Continue; 18 enabled Continue and tapping it opened 'Your height and weight'.
Status	Pass
Priority / Severity	High / Major
Tester / Execution date	Usman Ghani · 11-Oct-2026
Evidence	TC-003_age_119, TC-003_age18_body_step
Defects	—

Test Case ID	TC-004
Module / Feature	M1 Onboarding – Profile Entry
Linked requirement(s)	FR-003
Title / Description	Age 17 ends onboarding on the refusal screen
Technique / Type	BVA · Negative
Preconditions	Onboarding on the age step.
Test steps	1. Type 17. 2. Tap Continue.
Test data	Age = 17
Expected result	'Calori is built for adults' screen is shown with advice to speak to a doctor or guardian; no Continue button and no back arrow.
Actual result	Age 17 + Continue opened 'Calori is built for adults' with advice to talk to a doctor, school nurse, parent or guardian. No Continue button, no back arrow, progress bar hidden.
Status	Pass
Priority / Severity	High / Critical
Tester / Execution date	Usman Ghani · 11-Oct-2026
Evidence	TC-004_age17_blocked
Defects	—

Test Case ID	TC-006
Module / Feature	M1 Onboarding – Profile Entry
Linked requirement(s)	FR-004
Title / Description	Valid metric height and weight proceed
Technique / Type	EP · Positive
Preconditions	Onboarding on 'Your height and weight' (cm and kg selected).
Test steps	1. Type 178 in Height. 2. Type 78 in Current weight. 3. Tap Continue.
Test data	Height 178 cm, weight 78 kg
Expected result	Continue is enabled and opens 'What are you aiming for?' with the target pre-filled as 78 kg.
Actual result	178 cm + 78 kg enabled Continue, which opened 'What are you aiming for?' with Target weight pre-filled as 78 kg (note: the target field did not take focus by itself - see exploratory note).
Status	Pass
Priority / Severity	High / Major
Tester / Execution date	Usman Ghani · 11-Oct-2026
Evidence	TC-006_target_prefilled
Defects	—

Test Case ID	TC-010
Module / Feature	M1 Onboarding – Profile Entry
Linked requirement(s)	FR-005
Title / Description	Unit toggles convert typed values, independently
Technique / Type	EP · Positive
Preconditions	On the height/weight step, metric units.
Test steps	1. Type 178 cm and 78 kg. 2. Tap ft'in. 3. Tap lb. 4. Tap cm again.
Test data	178 cm, 78 kg
Expected result	ft'in shows 5 ft 10 in while weight stays in kg; lb shows 172 lb; switching height back shows 178 cm. Each toggle changes only its own measurement.
Actual result	178 cm -> ft'in showed 5 ft 10 in with weight still 78 kg; lb converted 78 kg to 172 lb; cm restored 178 cm while weight stayed 172 lb. Each toggle changed only its own measurement.
Status	Pass
Priority / Severity	Medium / Minor
Tester / Execution date	Usman Ghani · 11-Oct-2026
Evidence	TC-010_ftin_converted, TC-010_lb_converted, TC-010_back_to_cm
Defects	—

Test Case ID	TC-013
Module / Feature	M1 Onboarding – Profile Entry
Linked requirement(s)	FR-003
Title / Description	Under-18 refusal persists after the app is restarted
Technique / Type	EG / ST · Negative
Preconditions	Age 17 was entered and the refusal screen is showing.
Test steps	1. Force-stop Calori. 2. Relaunch it.
Test data	Age 17 previously entered
Expected result	The refusal remains in force (refusal screen shown again); the flow does not restart so that a different age can be typed.
Actual result	After force-stop and relaunch the app opened at step 1 of onboarding again, not the refusal screen. Entering age 25 enabled Continue and led to 'Your height and weight' - the under-18 stop is bypassed by restarting the app.
Status	Fail
Priority / Severity	High / Major
Tester / Execution date	Usman Ghani · 11-Oct-2026
Evidence	TC-013_after_restart, TC-013_reentered_age25
Defects	DEF-02

Test Case ID	TC-014
Module / Feature	M2 Goal Engine & Safety Rules
Linked requirement(s)	FR-007, FR-014
Title / Description	Reference profile produces the published target and macros
Technique / Type	EP (business rule) · Positive
Preconditions	Onboarding completed up to the target step for the reference profile.
Test steps	1. Enter target 72 kg, keep 12 weeks; note the inline note. 2. Continue; choose 'Moderately active'. 3. Read the plan screen.
Test data	Male, age 30, 178 cm, 78 kg, target 72 kg, 12 weeks, moderately active. Oracle: BMR 1747.5, TDEE 2708.6, 0.5 kg/week
Expected result	Inline note 'about 0.6% of your bodyweight a week'. Plan: 2,160 kcal; Protein 140 g, Carbs 265 g, Fat 60 g; 'reach your target around 3 January' (2027).
Actual result	Inline note 'That works out at about 0.6% of your bodyweight a week.' Plan: DAILY TARGET 2,160; Protein 140 g, Carbs 265 g, Fat 60 g; 'At this pace you reach your target around 3 January.' All match the oracle.
Status	Pass
Priority / Severity	High / Major
Tester / Execution date	Usman Ghani · 11-Oct-2026
Evidence	TC-014_target_note, TC-014_activity_step, TC-014_plan
Defects	—

Test Case ID	TC-015
Module / Feature	M2 Goal Engine & Safety Rules
Linked requirement(s)	FR-010
Title / Description	Too-fast loss rate is clamped to 1 %/week
Technique / Type	DT · Negative
Preconditions	Male, age 30, 178 cm, 78 kg; on the target step.
Test steps	1. Enter target 60 kg (12 weeks). 2. Read the inline message. 3. Continue, choose Moderately active and read the plan.
Test data	78 → 60 kg in 12 weeks = 1.5 kg/week > 0.78 kg/week cap. Oracle: 1,850 kcal, 22 March 2027
Expected result	Correction explains that more than 1 % a week is too fast and gives the honest date '22 March'; plan shows 1,850 kcal and repeats the explanation; Continue is allowed.
Actual result	Inline correction: 'That timeframe would mean losing more than 1% of your bodyweight a week, which is faster than is safe to keep up. At a steady rate you'd reach 60 kg by 22 March.' Continue enabled. Plan: 1,850 kcal, Protein 140 g / Carbs 208 g / Fat 51 g, explanation repeated - matches oracle.
Status	Pass
Priority / Severity	High / Major
Tester / Execution date	Usman Ghani · 11-Oct-2026
Evidence	TC-015_rate_clamped, TC-015_plan
Defects	—

Test Case ID	TC-016
Module / Feature	M2 Goal Engine & Safety Rules
Linked requirement(s)	FR-008
Title / Description	Target below BMI 18.5 is refused; 'Maintain instead' works
Technique / Type	DT · Negative
Preconditions	Male, age 30, 178 cm, 78 kg; on the target step.
Test steps	1. Enter target 55 kg. 2. Read the refusal and check Continue. 3. Tap 'Maintain instead', choose Moderately active.
Test data	55 kg at 178 cm = BMI 17.4; lowest healthy weight ≈ 58.6 kg. Oracle maintenance: 2,710 kcal
Expected result	Refusal states the target is below a healthy weight and names about 59 kg; Continue is disabled. 'Maintain instead' leads to a plan of 2,710 kcal, 'Enough to hold your current weight.'
Actual result	Refusal: 'That target is below a healthy weight for your height. The lowest weight in the healthy range for you is about 59 kg.' Continue disabled. 'Maintain instead' -> activity -> plan 2,710 kcal, 'Enough to hold your current weight.'
Status	Pass
Priority / Severity	High / Critical
Tester / Execution date	Usman Ghani · 11-Oct-2026
Evidence	TC-016_refused_55, TC-016_maintain_plan
Defects	—

Test Case ID	TC-018
Module / Feature	M2 Goal Engine & Safety Rules
Linked requirement(s)	FR-008
Title / Description	Underweight user's gain to a still-underweight target is offered a healthy-weight gain
Technique / Type	DT · Negative
Preconditions	Male, 30, 178 cm, current weight 52 kg; on the target step.
Test steps	1. Enter target 56 kg. 2. Read the message. 3. Tap 'Aim for a healthy weight instead', choose Moderately active, read the plan.
Test data	52 kg (BMI 16.4) → 56 kg (BMI 17.7). Healthy minimum ≈ 58.6 kg; maintenance 2,310 kcal
Expected result	Refusal with button 'Aim for a healthy weight instead'. The resulting plan is a weight-GAIN plan toward about 59 kg (daily target above maintenance, about 2,880 kcal, with a target date).
Actual result	Refusal shown with 'Aim for a healthy weight instead' (correct). Tapping it and choosing 'Moderately active' opened the plan screen showing ONLY an endless loading spinner - no target, no 'Start tracking', no back arrow. Still spinning after 15 s; system Back closes the app. Onboarding cannot be completed this way.
Status	Fail
Priority / Severity	High / Major
Tester / Execution date	Usman Ghani · 11-Oct-2026
Evidence	TC-018_refused_gain, TC-018_plan_after_healthy_weight, TC-018_spinner_after_15s
Defects	DEF-06

Test Case ID	TC-019
Module / Feature	M2 Goal Engine & Safety Rules
Linked requirement(s)	FR-010, FR-013
Title / Description	Too-fast GAIN rate is clamped with gain wording
Technique / Type	DT · Negative
Preconditions	Male, 30, 178 cm, 70 kg; on the target step.
Test steps	1. Enter target 85 kg (12 weeks). 2. Read the inline message. 3. Continue, Moderately active, read the plan.
Test data	70 → 85 kg in 12 weeks = 1.25 kg/week > 0.70 cap. Oracle: 3,360 kcal, 10 March 2027
Expected result	Correction says gaining more than 1 % a week is too fast and gives '10 March'; plan 3,360 kcal. The message must talk about gaining, not losing.
Actual result	Figures correct (3,360 kcal; 10 March; Protein 125 / Carbs 506 / Fat 93 g) but the explanation on the target step and plan reads 'That timeframe would mean LOSING more than 1% of your bodyweight a week…' for a weight-GAIN goal (70 -> 85 kg).
Status	Fail
Priority / Severity	Medium / Minor
Tester / Execution date	Usman Ghani · 11-Oct-2026
Evidence	TC-019_gain_too_fast, TC-019_plan
Defects	DEF-07

Test Case ID	TC-020
Module / Feature	M2 Goal Engine & Safety Rules
Linked requirement(s)	FR-011
Title / Description	Loss target below the female floor is clamped to 1,200 kcal
Technique / Type	DT / BVA · Negative
Preconditions	Female, 40, 155 cm, 60 kg; on the target step.
Test steps	1. Enter target 54 kg (12 weeks). 2. Continue, choose 'Not very active'. 3. Read the plan.
Test data	TDEE 1,449 − 550 = 899 < 1,200 floor. Oracle: 1,200 kcal, 15 April 2027
Expected result	Plan shows 1,200 kcal with the explanation that the date would put you under 1,200 kcal a day and that you would reach 54 kg by 15 April instead.
Actual result	Plan 1,200 kcal: 'Reaching that date would put you under 1200 kcal a day, which is lower than Calori will suggest. Holding at the floor, you'd reach 54 kg by 15 April instead.' Macros 110/109/36 g - matches oracle.
Status	Pass
Priority / Severity	High / Major
Tester / Execution date	Usman Ghani · 11-Oct-2026
Evidence	TC-020_floor_plan
Defects	—

Test Case ID	TC-021
Module / Feature	M2 Goal Engine & Safety Rules
Linked requirement(s)	FR-011
Title / Description	Maintenance already at or below the floor sets the plan to maintain
Technique / Type	DT (edge) · Edge
Preconditions	Female, 80, 150 cm, 45 kg; on the target step.
Test steps	1. Enter target 42 kg. 2. Continue, choose 'Not very active'. 3. Read the plan.
Test data	TDEE 992 kcal ≤ 1,200 floor
Expected result	Plan says your body already uses about 992 kcal a day, at or below the 1,200 minimum, so it is set to maintain; target 990 kcal; 'Enough to hold your current weight.'
Actual result	Plan 990 kcal, 'Enough to hold your current weight.' with 'Your body already uses about 992 kcal a day, which is at or below the 1200 kcal minimum… so this is set to maintain for now.'
Status	Pass
Priority / Severity	Medium / Major
Tester / Execution date	Usman Ghani · 11-Oct-2026
Evidence	TC-021_tdee_below_floor_plan
Defects	—

Test Case ID	TC-022
Module / Feature	M2 Goal Engine & Safety Rules
Linked requirement(s)	FR-012
Title / Description	Maintenance band boundary ±0.5 kg
Technique / Type	BVA · Edge
Preconditions	Male, age 30, 178 cm, 78 kg; on the target step.
Test steps	1. Enter target 78.5 and read the note. 2. Change to 78.6 and read the note.
Test data	78.5 kg (= +0.5) and 78.6 kg (> +0.5)
Expected result	78.5: 'Maintaining where you are. Calori will aim for about 2710 kcal a day.' 78.6: treated as a gain ('about 0.1% of your bodyweight a week').
Actual result	78.5 kg: 'Maintaining where you are. Calori will aim for about 2710 kcal a day.' 78.6 kg: 'That works out at about 0.1% of your bodyweight a week.' (gain).
Status	Pass
Priority / Severity	Medium / Minor
Tester / Execution date	Usman Ghani · 11-Oct-2026
Evidence	TC-022_78_5_maintain, TC-022_78_6_gain
Defects	—

Test Case ID	TC-025
Module / Feature	M2 Goal Engine & Safety Rules
Linked requirement(s)	FR-014, FR-001
Title / Description	'Start tracking' saves the profile and opens Home (survives restart)
Technique / Type	ST · Positive
Preconditions	Plan screen for the reference profile (2,160 kcal).
Test steps	1. Tap 'Start tracking'. 2. Force-stop and relaunch the app.
Test data	Reference plan
Expected result	Home opens showing '0 / 2,160 kcal' and 'nothing logged yet', protein 0 / 140 g. After relaunch Home is shown again, not onboarding.
Actual result	'Start tracking' opened Home: ring 0%, '0 / 2,160 kcal', 'nothing logged yet', PROTEIN 0 / 140 g, Carbs 0 g, Fat 0 g, 'Nothing logged yet.' After force-stop and relaunch Home was shown again with the same figures.
Status	Pass
Priority / Severity	High / Critical
Tester / Execution date	Usman Ghani · 11-Oct-2026
Evidence	TC-025_home_after_start, TC-025_home_after_relaunch
Defects	—

Test Case ID	TC-027
Module / Feature	M3 Manual Food Logging
Linked requirement(s)	FR-016
Title / Description	Regional names find English foods (channa, karela)
Technique / Type	EP · Positive
Preconditions	Add food open.
Test steps	1. Search channa. 2. Search karela.
Test data	Queries: channa, karela
Expected result	channa returns chickpea foods; karela returns bitter gourd foods.
Actual result	'channa' returned chickpea foods: 'Chickpea Cooked (Channa) - FCT-PK', 'Chickpea Raw (Channa) - FCT-PK', 'Chickpeas curry (Safed channa curry) - INDB', plus channa dishes. 'karela' returned 'Gourd, karela - COFID', 'Bitter Gourd (Karela) - FCT-PK', karela curries.
Status	Pass
Priority / Severity	Medium / Minor
Tester / Execution date	Faizan Ali · 11-Oct-2026
Evidence	TC-027_channa, TC-027_karela
Defects	—

Test Case ID	TC-028
Module / Feature	M3 Manual Food Logging
Linked requirement(s)	FR-017, FR-015
Title / Description	Unknown query shows 'No matches'; × clears it
Technique / Type	EP · Negative
Preconditions	Add food open.
Test steps	1. Search xqzvbn. 2. Tap the × in the search field.
Test data	Query: xqzvbn
Expected result	'No matches' with advice to try a shorter word; × empties the field and the results.
Actual result	'xqzvbn' showed 'No matches' with 'Try a shorter or more general word - "rice" finds more than "basmati rice, cooked".' The x button emptied the field and removed the message.
Status	Pass
Priority / Severity	Medium / Minor
Tester / Execution date	Faizan Ali · 11-Oct-2026
Evidence	TC-028_no_matches, TC-028_cleared
Defects	—

Test Case ID	TC-030
Module / Feature	M3 Manual Food Logging
Linked requirement(s)	FR-018
Title / Description	Recent foods are shown before typing after foods were logged by hand
Technique / Type	ST · Positive
Preconditions	At least two foods have been added through Add food today.
Test steps	1. Open Add food with an empty search field. 2. Look for a 'Recent' section.
Test data	Previously logged by hand today: Rice cake (TC-033)
Expected result	A 'Recent' section lists the recently/frequently logged foods before anything is typed.
Actual result	After logging Rice cake by hand, clearing the search field showed an empty screen - no 'Recent' section at all. Manually logged foods never appear as recent/frequent suggestions.
Status	Fail
Priority / Severity	Medium / Minor
Tester / Execution date	Faizan Ali · 11-Oct-2026
Evidence	TC-030_no_recent_section
Defects	DEF-12

Test Case ID	TC-032
Module / Feature	M3 Manual Food Logging
Linked requirement(s)	FR-019
Title / Description	kcal, grams and macros scale with quantity and serving
Technique / Type	EP (business rule) · Positive
Preconditions	A result with household servings is expanded.
Test steps	1. Note grams and kcal at quantity 1. 2. Press + once (the stepper moves in 0.5 steps) and note the values. 3. Choose a different serving chip. 4. Set the quantity to 0.5 and then to 20 and compare the values.
Test data	Quantity 1 → 1.5; second serving size; 0.5 → 20
Expected result	Grams and kcal scale exactly with the quantity (1.5× at 1.5, 40× from 0.5 to 20, ±1 for rounding); another serving changes grams and kcal in proportion; macros follow.
Actual result	Rice cake: 1 serving = 18 g / 71 kcal; one step up (1.5) = 27 g / 106 kcal (x1.5); serving '1 cake' x1.5 = 5 g / 18 kcal; 0.5 -> 20 cakes went 2 g / 6 kcal -> 60 g / 235 kcal. Grams, kcal and macros scale linearly.
Status	Pass
Priority / Severity	High / Major
Tester / Execution date	Faizan Ali · 11-Oct-2026
Evidence	TC-032_qty2, TC-032_one_cake_qty2
Defects	—

Test Case ID	TC-033
Module / Feature	M3 Manual Food Logging
Linked requirement(s)	FR-020, FR-021
Title / Description	'Add to log' saves to Home with a time-based meal type
Technique / Type	ST · Positive
Preconditions	Home on today; a result expanded; device time 18:00–22:00.
Test steps	1. Tap 'Add to log'. 2. Observe the snackbar and Home.
Test data	Rice cake, 20 × '1 cake' (60 g, 235 kcal) as left in the picker by TC-032; time 20:46
Expected result	Snackbar '<food> added'; Home shows a 'Dinner' card with the food's kcal; ring and totals increase by that amount.
Actual result	Snackbar 'Rice cake added'; returned to Home: DINNER card 'Rice cake 235 kcal' (logged at 20:46), ring 11%, '235 / 2,160 kcal left today', Protein 4 / 140 g, Carbs 49 g, Fat 3 g.
Status	Pass
Priority / Severity	High / Critical
Tester / Execution date	Faizan Ali · 11-Oct-2026
Evidence	TC-033_snackbar, TC-033_home_after_add
Defects	—

Test Case ID	TC-034
Module / Feature	M3 Manual Food Logging
Linked requirement(s)	FR-020
Title / Description	A second food can be added after returning to Add food
Technique / Type	ST / EG · Positive
Preconditions	One food was just added through Add food (TC-033).
Test steps	1. Tap '+' again. 2. Observe the previously used result card. 3. Tap 'Add to log' on it (or expand and add).
Test data	Same session, same query
Expected result	The card is back in its resting state with 'Add to log' enabled; the second add saves a new entry and returns to Home.
Actual result	Returning to Add food showed the same Rice cake card still expanded with its button reading 'Saving…' and disabled. Tapping it did nothing; Home still had only the one entry. The card stays stuck for the rest of the session, so that food cannot be added again from it.
Status	Fail
Priority / Severity	High / Major
Tester / Execution date	Faizan Ali · 11-Oct-2026
Evidence	TC-034_stuck_saving, V1_stuck_saving_button.mp4
Defects	DEF-11

Test Case ID	TC-036
Module / Feature	M4 Day View & Entry Editing
Linked requirement(s)	FR-021
Title / Description	Ring, calorie line and macros reflect logged food
Technique / Type	EP (business rule) · Positive
Preconditions	Today has known entries; target 2,160 kcal.
Test steps	1. Read the ring %, the 'X / Y kcal' line, the protein card and the meal cards.
Test data	Entries: Rice cake 235 kcal + Boiled egg (Ubla anda) 68 kcal
Expected result	Ring % = round(consumed ÷ 2,160 × 100); line '<consumed> / 2,160 kcal' 'left today'; protein/carbs/fat equal the sums of the entries.
Actual result	Entries 235 + 68 kcal: ring 14%, '304 / 2,160 kcal left today' (cards round individually; exact sum 303.8), Protein 11 / 140 g, Carbs 49 g, Fat 7 g - consistent with the entries.
Status	Pass
Priority / Severity	High / Major
Tester / Execution date	Faizan Ali · 11-Oct-2026
Evidence	TC-036_home_two_entries
Defects	—

Test Case ID	TC-037
Module / Feature	M4 Day View & Entry Editing
Linked requirement(s)	FR-022
Title / Description	Over target shown neutrally, never negative
Technique / Type	BVA · Edge
Preconditions	Today's total is below target.
Test steps	1. Add a large portion so the day exceeds 2,160 kcal. 2. Read Home.
Test data	10 cups of 'Biryani with chicken' (USDA) = 2,038 kcal
Expected result	Line reads '<consumed> / 2,160 kcal' with 'over today'; ring above 100 %; no negative number and no red.
Actual result	After adding 10 cups of Biryani with chicken (2,038 kcal): ring 105% with a dim second lap, '2,274 / 2,160 kcal' 'over today'; no negative number, no red. (Side note: the '+' button covers the last card's kcal - exploratory note.)
Status	Pass
Priority / Severity	Medium / Minor
Tester / Execution date	Faizan Ali · 11-Oct-2026
Evidence	TC-037_over_target
Defects	—

Test Case ID	TC-038
Module / Feature	M4 Day View & Entry Editing
Linked requirement(s)	FR-023
Title / Description	Date strip shows 12 days ending today, no future days
Technique / Type	BVA · Positive
Preconditions	Home on today.
Test steps	1. Scroll the date strip to its oldest end. 2. Count the days and check the newest.
Test data	Today = 11 Oct 2026
Expected result	Exactly 12 chips from 30 Sep to 11 Oct; 11 Oct is the newest; no 12 Oct or later.
Actual result	Strip holds exactly 12 chips: Wed 30 Sep … Sun 11 Oct; 11 Oct is newest and no future dates are offered.
Status	Pass
Priority / Severity	Medium / Minor
Tester / Execution date	Faizan Ali · 11-Oct-2026
Evidence	TC-038_strip_oldest, TC-038_strip_newest
Defects	—

Test Case ID	TC-039
Module / Feature	M4 Day View & Entry Editing
Linked requirement(s)	FR-024
Title / Description	Changing an entry's meal type in the editor
Technique / Type	ST · Positive
Preconditions	A Dinner entry exists today.
Test steps	1. Tap the entry. 2. Tap 'Breakfast'. 3. Close the sheet.
Test data	Dinner → Breakfast
Expected result	The chip selection moves to Breakfast immediately and the Home card's label reads Breakfast.
Actual result	In the editor, tapping Breakfast moved the selection immediately; after closing, the Home card read 'BREAKFAST - Boiled egg (Ubla anda)'.
Status	Pass
Priority / Severity	High / Major
Tester / Execution date	Faizan Ali · 11-Oct-2026
Evidence	TC-039_editor_open, TC-039_breakfast_selected, TC-039_home_breakfast_label
Defects	—

Test Case ID	TC-044
Module / Feature	M4 Day View & Entry Editing
Linked requirement(s)	FR-025
Title / Description	'Remove this meal' deletes the entry and closes the editor
Technique / Type	ST · Positive
Preconditions	Entry editor open.
Test steps	1. Tap 'Remove this meal'.
Test data	—
Expected result	The sheet closes, the entry is gone and totals drop; the app keeps working (no blank screen).
Actual result	'Remove this meal' closed the editor; 'Biryani with chicken' disappeared and Home returned to 235 / 2,160 kcal (11%). App kept working normally - no blank screen.
Status	Pass
Priority / Severity	Medium / Major
Tester / Execution date	Faizan Ali · 11-Oct-2026
Evidence	TC-044_editor, TC-044_after_remove
Defects	—

Test Case ID	TC-045
Module / Feature	M4 Day View & Entry Editing
Linked requirement(s)	FR-026
Title / Description	Home updates when the calorie target is changed
Technique / Type	ST · Positive
Preconditions	Today has entries; Home shows ring % against 2,160 kcal.
Test steps	1. Note the ring % and calorie line. 2. On Goal, set daily calories by hand to half the previous target. 3. Return to Home.
Test data	Target 2,160 → 1,080 kcal
Expected result	Both the calorie line and the ring % reflect the new target immediately (ring % doubles).
Actual result	Before: ring 11%, '235 / 2,160 kcal'. After setting 1,080 kcal on Goal: line '235 / 1,080 kcal' but the ring still showed 11% (should be 22%). It corrected to 22% only after selecting another day and coming back. The calendar also kept the old target until the app was restarted.
Status	Fail
Priority / Severity	High / Major
Tester / Execution date	Faizan Ali · 11-Oct-2026
Evidence	TC-045_home_before, TC-045_home_after_goal_change, TC-045_today_after_day_switch
Defects	DEF-21

Test Case ID	TC-046
Module / Feature	M5 Calendar
Linked requirement(s)	FR-027, FR-045
Title / Description	Calendar opens with rings for logged days and dots for unlogged days
Technique / Type	EP · Positive
Preconditions	Entries exist today and yesterday.
Test steps	1. On Home tap the calendar button. 2. Inspect October 2026.
Test data	October 2026
Expected result	'October 2026' grid, Monday first; days with entries drawn as rings; unlogged past days with a neutral dot; today outlined; future days dimmed; legend shown.
Actual result	Calendar opened on 'October 2026', Monday-first; 10 and 11 Oct drawn as small rings (3% and 11% of target), unlogged past days with a neutral dot, today outlined, 12-31 Oct dimmed; legend 'on target / over / not logged'; forward arrow greyed.
Status	Pass
Priority / Severity	High / Major
Tester / Execution date	Faizan Ali · 11-Oct-2026
Evidence	TC-046_calendar_october
Defects	—

Test Case ID	TC-048
Module / Feature	M5 Calendar
Linked requirement(s)	FR-028
Title / Description	Paging months: back works, forward stops at the current month
Technique / Type	BVA / ST · Positive
Preconditions	Calendar on October 2026.
Test steps	1. Tap the forward arrow. 2. Tap the back arrow. 3. Tap forward again.
Test data	Current month = October 2026
Expected result	Forward is disabled on October; back shows September 2026; forward then returns to October and is disabled again.
Actual result	Forward arrow did nothing on October 2026; back showed 'September 2026'; forward returned to October and was inert again.
Status	Pass
Priority / Severity	Medium / Minor
Tester / Execution date	Faizan Ali · 11-Oct-2026
Evidence	TC-048_september
Defects	—

Test Case ID	TC-049
Module / Feature	M5 Calendar
Linked requirement(s)	FR-029
Title / Description	Day detail shows consumed vs target, macros and entries
Technique / Type	ST · Positive
Preconditions	Calendar on October with today logged.
Test steps	1. Tap today. 2. Read the detail card. 3. Tap today again.
Test data	Today
Expected result	Detail lists '<kcal> of 2,160 kcal', 'within/over target', protein, carbs and fat, and each entry; a second tap closes it.
Actual result	Tapping 11 Oct showed '11 OCTOBER - 235 of 2,160 kcal - within target - Protein 4 g - Carbs 49 g - Rice cake 235'; a second tap closed it (correct). Fat is missing from the day detail, although Home shows it.
Status	Fail
Priority / Severity	Medium / Minor
Tester / Execution date	Faizan Ali · 11-Oct-2026
Evidence	TC-049_day_detail, TC-049_closed
Defects	DEF-17

Test Case ID	TC-051
Module / Feature	M5 Calendar
Linked requirement(s)	FR-030
Title / Description	Month summary averages over logged days only
Technique / Type	EP (business rule) · Positive
Preconditions	Two days logged in October with known totals; no day selected.
Test steps	1. Read the 'Average' and 'Logged' cards.
Test data	Day totals from Home
Expected result	Average = (sum of logged days' kcal) ÷ (number of logged days), rounded; Logged = number of days with entries.
Actual result	Logged days 10 Oct (72 kcal) and 11 Oct (235 kcal): 'AVERAGE 153 kcal per logged day' ((72+235)/2 = 153.5, exact values 153.x), 'LOGGED 2 days this month'.
Status	Pass
Priority / Severity	Medium / Minor
Tester / Execution date	Faizan Ali · 11-Oct-2026
Evidence	TC-046_calendar_october
Defects	—

Test Case ID	TC-052
Module / Feature	M5 Calendar
Linked requirement(s)	FR-031
Title / Description	A past day keeps the target it was logged against
Technique / Type	DT (business rule) · Negative
Preconditions	Yesterday logged against 2,160 kcal; note its ring % in Calendar and Home.
Test steps	1. On Goal, set calories by hand to 1,080 (done in TC-045). 2. Reopen yesterday on Home, and in Calendar (also after a restart).
Test data	Target 2,160 → 1,080 kcal
Expected result	Yesterday still reads 'of 2,160 kcal' with the same ring %; only today onward uses 1,080 (the Goal screen promises this).
Actual result	Yesterday (10 Oct, 72 kcal) was logged against 2,160 kcal (3%). After changing the goal to 1,080 kcal, Home shows yesterday as '72 / 1,080 kcal' 7%, and after a restart the calendar shows '10 OCTOBER - 72 of 1,080 kcal', 7%. Past days are re-judged against the new target, contradicting the Goal screen's 'Days you have already logged keep the target they were measured against'.
Status	Fail
Priority / Severity	High / Major
Tester / Execution date	Faizan Ali · 11-Oct-2026
Evidence	TC-052_yesterday_home_after_change, TC-052_calendar_yesterday, TC-052_calendar_yesterday_after_restart
Defects	DEF-22

Test Case ID	TC-053
Module / Feature	M6 Goal Management
Linked requirement(s)	FR-032
Title / Description	Goal screen shows the saved plan
Technique / Type	EP · Positive
Preconditions	Reference profile saved (2,160 kcal).
Test steps	1. Tap the Goal tab. 2. Read the card and steppers.
Test data	Reference profile
Expected result	2,160 kcal; Protein 140 g, Carbs 265 g, Fat 60 g; caption 'On track to reach your target by 3 January'; current 78 kg, goal 72 kg, rate about 0.6 %.
Actual result	Goal screen: DAILY TARGET 2,160 (pencil), 'On track to reach your target by 3 January.', Protein 140 g / Carbs 265 g / Fat 60 g, Current weight 78 kg, Goal weight 72 kg, Rate of loss 0.6 %.
Status	Pass
Priority / Severity	High / Major
Tester / Execution date	Faizan Ali · 11-Oct-2026
Evidence	TC-053_goal_screen
Defects	—

Test Case ID	TC-056
Module / Feature	M6 Goal Management
Linked requirement(s)	FR-033, FR-007
Title / Description	Target honours the chosen rate (95 → 92 kg at 1.0 %/week)
Technique / Type	EP / BVA (business rule) · Positive
Preconditions	Profile: Male, 30, 178 cm, 95 kg, goal 92 kg, moderately active; Goal screen open.
Test steps	1. Tap + on the rate until it reads 1.0 %. 2. Read the target and caption.
Test data	Oracle at 0.95 kg/week: 1,930 kcal, reached 3 November
Expected result	Target 1,930 kcal; caption 'At 1.0 % per week you reach your target by 3 November'.
Actual result	Rate stepper shows 1.0 % but the card says 'At 0.8 % per week you reach your target by 8 November' and 2,150 kcal. At the chosen 1.0 % (0.95 kg/week) the target should be 1,930 kcal by 3 November - the screen sets a target 220 kcal higher than the rate the user picked.
Status	Fail
Priority / Severity	High / Major
Tester / Execution date	Faizan Ali · 11-Oct-2026
Evidence	TC-056_rate_1_0_quantised
Defects	DEF-24

Test Case ID	TC-057
Module / Feature	M6 Goal Management
Linked requirement(s)	FR-034
Title / Description	Unsafe goal weight is refused on the Goal screen and not saved
Technique / Type	DT · Negative
Preconditions	Goal screen after TC-056 (current 95 kg, goal 92 kg).
Test steps	1. Step Goal weight down to 58 kg. 2. Read the card. 3. Force-stop, relaunch, open Goal.
Test data	58 kg at 178 cm = BMI 18.3
Expected result	The card shows the refusal (lowest healthy weight about 59 kg) and no target; after relaunch the last safe goal is still saved.
Actual result	At 58 kg the card showed '-' for the target and 'That target is below a healthy weight… about 59 kg.' After force-stop and relaunch, Goal weight was back to the last safe value (92 kg) and the refused goal was not saved.
Status	Pass
Priority / Severity	High / Major
Tester / Execution date	Faizan Ali · 11-Oct-2026
Evidence	TC-057_refused_58, TC-057_after_restart
Defects	—

Test Case ID	TC-058
Module / Feature	M6 Goal Management
Linked requirement(s)	FR-035
Title / Description	Hand-set calories: suggestion, floor advisory boundary and Reset
Technique / Type	ST / BVA · Positive
Preconditions	Goal screen, male profile (floor 1,500).
Test steps	1. Tap the kcal figure, enter 1500, Save. 2. Repeat with 1499. 3. Tap 'Reset'.
Test data	1500 (= floor), 1499 (floor − 1)
Expected result	1500: target 1,500 with 'Calori suggests 2,160 · Reset', no advisory. 1499: advisory 'below the 1500 kcal a day…'. Reset returns to 2,160 and hides both lines.
Actual result	1500 -> target 1,500 with 'Calori suggests 2,160 - Reset', no advisory. 1499 -> adds 'That is below the 1500 kcal a day Calori would suggest as a minimum. Your call…' (macros 140/129/47 g = 1,499 kcal). Reset -> back to 2,160, both lines gone.
Status	Pass
Priority / Severity	Medium / Minor
Tester / Execution date	Faizan Ali · 11-Oct-2026
Evidence	TC-058_kcal_sheet, TC-058_kcal_1500, TC-058_kcal_1499_advisory, TC-058_after_reset
Defects	—

Test Case ID	TC-061
Module / Feature	M6 Goal Management
Linked requirement(s)	FR-036
Title / Description	Changing the weight unit keeps the goal and any hand-set values
Technique / Type	EG / ST · Negative
Preconditions	Goal screen with a loss goal and calories set by hand to 2,000 (suggestion line visible).
Test steps	1. Note caption, rate and kcal. 2. Tap 'lb' in About → Weight. 3. Read the card and steppers.
Test data	kg → lb
Expected result	Steppers switch to pounds; the caption, rate, target and the hand-set 2,000 kcal with its 'Calori suggests' line are unchanged.
Actual result	Before: 'Calori suggests 2,160 - Reset', hand-set 2,000 kcal, 'On track to reach your target by 3 January.', rate 0.6 %. After tapping lb: caption became 'Enough to hold your current weight.' although the goal is still 172 -> 159 lb, rate jumped to 0.5 %, suggestion became 2,300. After a restart the 'Calori suggests… Reset' line was gone too: the toggle had erased the saved target date and the hand-set flag.
Status	Fail
Priority / Severity	High / Major
Tester / Execution date	Faizan Ali · 11-Oct-2026
Evidence	TC-061_before_units_toggle, TC-061_after_lb, TC-061_after_restart, V2_units_toggle_wipes_goal.mp4
Defects	DEF-23

Test Case ID	TC-062
Module / Feature	M6 Goal Management
Linked requirement(s)	FR-037
Title / Description	Sources screen lists data sources and returns
Technique / Type	EP · Positive
Preconditions	Goal screen.
Test steps	1. Tap 'Where the food data comes from'. 2. Read the list. 3. Go back.
Test data	—
Expected result	Sources screen lists the food databases (USDA, CNF, CoFID, CIQUAL, INDB, Pakistan FCT) with attributions; back returns to Goal.
Actual result	Sources screen ('Food data') lists CIQUAL, Canadian Nutrient File, Food Composition Table for Pakistan, INDB, CoFID, USDA FNDDS, USDA Foundation and USDA SR Legacy, each with its attribution text; system Back returned to Goal.
Status	Pass
Priority / Severity	Low / Minor
Tester / Execution date	Faizan Ali · 11-Oct-2026
Evidence	TC-062_sources, TC-062_sources_2
Defects	—

Test Case ID	TC-063
Module / Feature	M7 Photo Logging
Linked requirement(s)	FR-038
Title / Description	Viewfinder opens; details field not focused; close returns
Technique / Type	EP · Positive
Preconditions	Home showing; build has photo estimates configured.
Test steps	1. Tap the camera button. 2. Observe the keyboard and the 'Add details' control. 3. Tap Close.
Test data	—
Expected result	Viewfinder with shutter, Gallery and 'Add details'; no keyboard appears by itself; Close returns to Home with nothing logged.
Actual result	Viewfinder opened (after a one-time camera permission prompt) with 'Add details' and the shutter; no keyboard appeared (correct). But the Close (x) button sits under the status bar (button y 32-147 px, status bar 0-142 px): two taps on it did nothing; only a tap in the 5 px strip at y=145 closed it. Also, no 'Gallery' label is visible: the Gallery pill stretches across the whole bar and its label is hidden behind the shutter.
Status	Fail
Priority / Severity	High / Major
Tester / Execution date	Usman Ghani · 11-Oct-2026
Evidence	TC-063_viewfinder, TC-063_after_close, EXP_camera_permission_prompt
Defects	DEF-26, DEF-25

Test Case ID	TC-064
Module / Feature	M7 Photo Logging
Linked requirement(s)	FR-039
Title / Description	Gallery meal photo → proposed items → Save logs an AI entry with photo
Technique / Type	ST · Positive
Preconditions	Viewfinder open; test images in the gallery.
Test steps	1. Tap Gallery, choose biryani.jpg. 2. Wait for the estimate; read the proposed items. 3. Tap 'Save to log'. 4. Open the new entry on Home.
Test data	biryani.jpg (CC0, Wikimedia Commons)
Expected result	Proposed items (e.g. biryani, naan) with kcal, grams and macros; nothing logged before Save; after Save 'Meal added', a new entry on Home whose editor shows the photo.
Actual result	Gallery -> biryani.jpg: 'WHAT WE FOUND 2,150 kcal' with paratha 850, mutton curry 600, biryani with egg 550, raita 80, sambar 70 kcal (grams, portion text, macros each). Nothing logged before Save. 'Save to log' -> 'Meal added' and a Dinner entry whose editor shows the photo (correct) - BUT the app returned to the live camera viewfinder instead of Home; the user had to press system Back to see the meal.
Status	Fail
Priority / Severity	High / Critical
Tester / Execution date	Usman Ghani · 11-Oct-2026
Evidence	TC-064_picker, TC-064_proposed, TC-064_proposed_2, TC-064_meal_added_snackbar, TC-064_home_after_save, TC-064_entry_with_photo
Defects	DEF-29

Test Case ID	TC-065
Module / Feature	M7 Photo Logging
Linked requirement(s)	FR-040
Title / Description	Details typed before choosing a gallery photo are applied
Technique / Type	EP · Positive
Preconditions	Viewfinder open.
Test steps	1. Tap 'Add details', type 'three apples, about 540 g', Done. 2. Tap Gallery, choose apple.jpg. 3. Read the proposed portion.
Test data	apple.jpg shows ONE apple; hint says three apples ≈ 540 g
Expected result	The proposal follows the user's details (about 3 apples / ≈540 g), not the single apple visible.
Actual result	Details 'three apples, about 540 g' were accepted (chip 'Details: three apples, about 540 g'), then Gallery -> apple.jpg. The proposal was 'apple - 95 kcal - 182 g - 1 medium apple': the typed details were ignored for a gallery photo.
Status	Fail
Priority / Severity	Medium / Minor
Tester / Execution date	Usman Ghani · 11-Oct-2026
Evidence	TC-065_details_typed, TC-065_details_set, TC-065_apple_proposal
Defects	DEF-30

Test Case ID	TC-066
Module / Feature	M7 Photo Logging
Linked requirement(s)	FR-041
Title / Description	Portion of a proposed item can be adjusted before saving
Technique / Type	EP · Positive
Preconditions	Review screen showing proposed items for a meal photo.
Test steps	1. Look for a portion control on each item. 2. Tap an item's portion text and try to change it.
Test data	Proposed items for biryani.jpg (review screen of TC-064, before saving)
Expected result	Every proposed item's portion can be changed (grams or serving multiplier) and its kcal and the total update live.
Actual result	On the biryani review none of the five items offered any portion control: tapping '250 g - 5 pieces' (and the other items) did nothing; there is no stepper, chip or grams field. Only the x (remove) is available, so a wrong portion cannot be corrected before saving.
Status	Fail
Priority / Severity	High / Major
Tester / Execution date	Usman Ghani · 11-Oct-2026
Evidence	TC-066_portion_tap_no_control
Defects	DEF-27

Test Case ID	TC-069
Module / Feature	M7 Photo Logging
Linked requirement(s)	FR-042
Title / Description	A photo with no food is reported as such
Technique / Type	EP · Negative
Preconditions	Viewfinder open.
Test steps	1. Tap Gallery, choose cat_nonfood.jpg. 2. Read the result.
Test data	cat_nonfood.jpg (CC0)
Expected result	'No food in that photo' with 'Try again' and 'Add by hand instead'; no items and nothing logged.
Actual result	cat_nonfood.jpg -> 'No food in that photo - Try again with the meal filling more of the frame, or add it by hand.' with 'Try again' and 'Add by hand instead'; no items, nothing logged.
Status	Pass
Priority / Severity	High / Major
Tester / Execution date	Usman Ghani · 11-Oct-2026
Evidence	TC-069_cat_no_food
Defects	—

Test Case ID	TC-070
Module / Feature	M7 Photo Logging
Linked requirement(s)	FR-043
Title / Description	Offline failure shows a specific message; retry keeps the photo
Technique / Type	EG / ST · Negative
Preconditions	Airplane mode on; viewfinder open.
Test steps	1. Choose salad.jpg from Gallery. 2. Read the message. 3. Turn airplane mode off; tap 'Try again'.
Test data	salad.jpg; airplane mode on, then off
Expected result	Message 'No connection, so the photo can't be analysed right now…' with Try again and 'Add by hand instead'; after reconnecting, Try again analyses the same photo without choosing it again.
Actual result	Airplane mode on -> salad.jpg -> 'No connection, so the photo can't be analysed right now. You can still add this meal by hand.' with Try again and Add by hand instead. Airplane mode off -> 'Try again' analysed the same photo without re-picking: 'mixed salad - 250 kcal - 300 g - 1 bowl'.
Status	Pass
Priority / Severity	High / Major
Tester / Execution date	Usman Ghani · 11-Oct-2026
Evidence	TC-070_offline_message, TC-070_retry_result
Defects	—

Test Case ID	TC-072
Module / Feature	M7 Photo Logging
Linked requirement(s)	FR-044
Title / Description	The same meal photographed again gives the same numbers
Technique / Type	EP · Positive
Preconditions	biryani.jpg was saved in TC-064.
Test steps	1. Choose biryani.jpg again from Gallery. 2. Compare the proposed figures with the saved entry.
Test data	biryani.jpg, second time
Expected result	The items and kcal match the first saved values (items marked as your saved value where applicable).
Actual result	biryani.jpg a second time gave the same five items with identical kcal, grams and macros, each marked 'Using the numbers you accepted before'.
Status	Pass
Priority / Severity	Medium / Minor
Tester / Execution date	Usman Ghani · 11-Oct-2026
Evidence	TC-072_biryani_repeat
Defects	—

Test Case ID	TC-074
Module / Feature	M8 Navigation
Linked requirement(s)	FR-045
Title / Description	Nav bar switches Home and Goal with the active tab highlighted
Technique / Type	ST · Positive
Preconditions	Home showing.
Test steps	1. Tap Goal. 2. Tap Home.
Test data	—
Expected result	Goal opens and its tab shows a highlighted pill with its label; Home returns and becomes the highlighted tab.
Actual result	Tapping Goal opened the Goal screen and its tab became selected (selected=true, highlighted pill with label); tapping Home returned to Home and Home became the selected tab.
Status	Pass
Priority / Severity	High / Major
Tester / Execution date	Faizan Ali · 11-Oct-2026
Evidence	TC-074_goal_tab, TC-074_home_tab
Defects	—

Test Case ID	TC-076
Module / Feature	M8 Navigation
Linked requirement(s)	FR-046
Title / Description	System Back from Calendar / Add food / Goal returns to Home
Technique / Type	EG · Negative
Preconditions	Calendar showing (opened from Home).
Test steps	1. Press the Android system Back button. 2. Repeat from Add food and from Goal.
Test data	—
Expected result	Each time the app returns to Home; it does not close or go to the background.
Actual result	System Back from Calendar, from Add food and from Goal each closed Calori (launcher came to the front) instead of returning to Home. Reopening showed Home.
Status	Fail
Priority / Severity	Medium / Minor
Tester / Execution date	Faizan Ali · 11-Oct-2026
Evidence	TC-076_back_from_calendar
Defects	DEF-03

Test Case ID	TC-077
Module / Feature	M2 Goal Engine & Safety Rules
Linked requirement(s)	FR-009, FR-014
Title / Description	'Maintain instead' for an underweight user produces a maintenance plan
Technique / Type	DT / EG · Negative
Preconditions	Male, 30, 178 cm, 55 kg (BMI 17.4); target 52 kg refused as a loss goal from an underweight user.
Test steps	1. Tap 'Maintain instead'. 2. Choose 'Moderately active'. 3. Wait 10 s and read the plan screen.
Test data	Oracle maintenance for 55 kg: TDEE 2,352 → 2,350 kcal
Expected result	A maintenance plan of about 2,350 kcal with 'Start tracking', exactly as 'Maintain instead' promises.
Actual result	After 'Maintain instead' and 'Moderately active' the plan screen shows only an endless loading spinner (still spinning after 10 s) - no target, no 'Start tracking', no back arrow. The offered alternative is a dead end, so an underweight user cannot finish onboarding.
Status	Fail
Priority / Severity	High / Critical
Tester / Execution date	Usman Ghani · 11-Oct-2026
Evidence	TC-017_maintain_spinner
Defects	DEF-06

Appendix D – Defect Log and Defect Reports

D.1 Defect log

Table 26: Defect log

ID	Title	Module	Severity	Priority	Status	Found by	Reporter	Date
DEF-02	Under-18 refusal is forgotten when the app is restarted	M1	Major	High	Open	TC-013	Usman Ghani	11-Oct-2026
DEF-03	System Back closes the app instead of going back (onboarding and secondary screens)	M8	Major	Medium	Open	TC-076	Faizan Ali	11-Oct-2026
DEF-05	Onboarding progress bar is full one step early	M1	Trivial	Low	Open	EXP-01	Usman Ghani	11-Oct-2026
DEF-06	Underweight users' offered alternatives end on an endless loading spinner	M2	Critical	High	Open	TC-018, TC-077	Usman Ghani	11-Oct-2026
DEF-07	'Too fast' correction says 'losing' on a weight-gain goal	M2	Minor	Medium	Open	TC-019	Usman Ghani	11-Oct-2026
DEF-11	'Add to log' stays stuck on 'Saving…' after the first manual add	M3	Major	High	Open	TC-034	Faizan Ali	11-Oct-2026
DEF-12	Recent foods never appear for foods logged by hand	M3	Minor	Medium	Open	TC-030	Faizan Ali	11-Oct-2026
DEF-13	Implausible nutrition values for some INDB foods	M3	Major	Medium	Open	EXP-02	Faizan Ali	11-Oct-2026
DEF-14	Portion chip labelled '0.8×' applies 0.75×	M4	Minor	Low	Open	EXP-03	Faizan Ali	11-Oct-2026
DEF-17	Calendar day detail omits fat	M5	Trivial	Low	Open	TC-049	Faizan Ali	11-Oct-2026
DEF-21	Home ring % (and Calendar) keep the old target after a goal change	M4	Major	High	Open	TC-045	Faizan Ali	11-Oct-2026
DEF-22	Past days are re-judged against a new goal	M5	Major	High	Open	TC-052	Faizan Ali	11-Oct-2026
DEF-23	Switching the weight unit erases the goal date and hand-set calories	M6	Major	High	Open	TC-061	Faizan Ali	11-Oct-2026
DEF-24	Goal screen ignores the chosen rate (target up to 220 kcal off)	M6	Major	High	Open	TC-056	Faizan Ali	11-Oct-2026
DEF-25	Viewfinder's Gallery button has no visible label	M7	Major	Medium	Open	TC-063	Usman Ghani	11-Oct-2026
DEF-26	Viewfinder's Close (×) sits under the status bar and cannot be tapped	M7	Major	High	Open	TC-063	Usman Ghani	11-Oct-2026
DEF-27	Proposed photo items' portions cannot be adjusted before saving	M7	Major	High	Open	TC-066	Usman Ghani	11-Oct-2026
DEF-29	After saving a gallery photo the app returns to the camera, not Home	M7	Minor	Medium	Open	TC-064	Usman Ghani	11-Oct-2026
DEF-30	Details typed before choosing a gallery photo are ignored	M7	Major	Medium	Open	TC-065	Usman Ghani	11-Oct-2026
DEF-33	Fields meant to take focus do not (Add food search, target weight)	M3	Trivial	Low	Open	EXP-01	Faizan Ali	11-Oct-2026
DEF-34	Past days say 'left today'	M4	Trivial	Low	Open	EXP-06	Faizan Ali	11-Oct-2026

D.2 Defect reports

One report per defect, with the device screenshot beside it; a red highlight was added to mark the fault. The unmarked originals and any further screenshots named under Evidence are in the evidence folder (Appendix F). Environment for every defect: Android 16 (API 36) emulator 'finall', 1080x2424 @ 420 dpi; release APK built from commit 35e2473.

DEF-02 – Under-18 refusal is forgotten when the app is restarted

Defect ID	DEF-02
Title	Under-18 refusal is forgotten when the app is restarted
Module / Feature	M1 Onboarding – Profile Entry
Found by	TC-013
Severity / Priority	Major / High
Status	Open
Steps to reproduce	1. Clear app data, launch, tap Male, enter age 17, tap Continue ('Calori is built for adults'). 2. Force-stop Calori and relaunch it. 3. Tap Male, enter age 25, tap Continue.
Expected result	The refusal stays in force after a restart (UC-02: 'No path forward').
Actual result	The app restarts onboarding at step 1; any adult age is then accepted and the flow continues to 'Your height and weight'.
Impact	The app's only hard safety stop can be bypassed in seconds by closing and reopening it.
Evidence	TC-013_after_restart, TC-013_reentered_age25
Reporter / Date	Usman Ghani · 11-Oct-2026

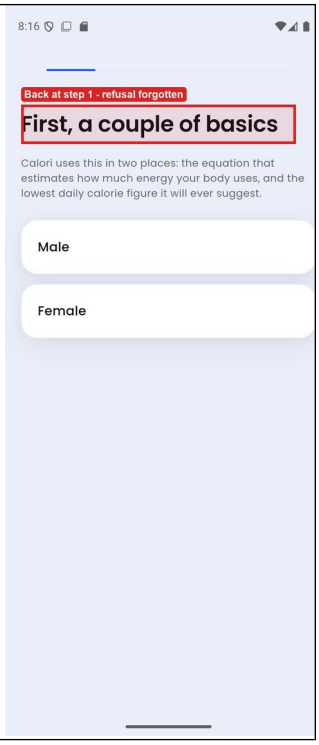


Figure 5: DEF-02 evidence: TC-013_after_restart (highlight added)

DEF-03 – System Back closes the app instead of going back (onboarding and secondary screens)

Defect ID	DEF-03
Title	System Back closes the app instead of going back (onboarding and secondary screens)
Module / Feature	M8 Navigation
Found by	TC-076
Severity / Priority	Major / Medium
Status	Open
Steps to reproduce	1. During onboarding, on 'Your height and weight' with the keyboard closed, press the Android Back button. 2. Or, with a profile saved, open Calendar (or Add food, or Goal) and press Back.
Expected result	Back returns to the previous onboarding step, or to Home from a secondary screen.
Actual result	Calori closes and the launcher appears. In onboarding every answer is lost: reopening starts again at step 1.
Impact	The most-used Android gesture exits the app and discards onboarding progress.
Evidence	TC-012_system_back_onboarding, TC-012_before_back, TC-076_back_from_calendar
Reporter / Date	Faizan Ali · 11-Oct-2026

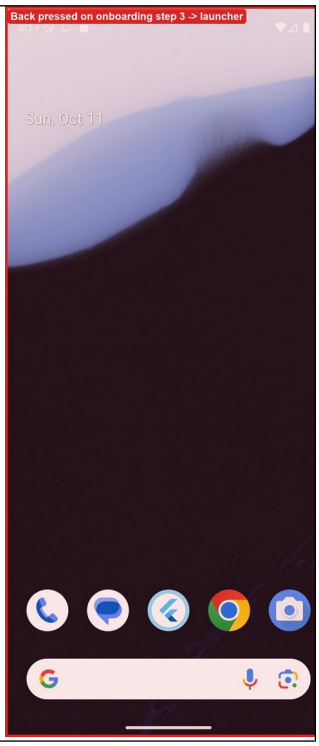


Figure 6: DEF-03 evidence: TC-012_system_back_onboarding (highlight added)

DEF-05 – Onboarding progress bar is full one step early

Defect ID	DEF-05
Title	Onboarding progress bar is full one step early
Module / Feature	M1 Onboarding – Profile Entry
Found by	EXP-01
Severity / Priority	Trivial / Low
Status	Open
Steps to reproduce	1. Go through onboarding to 'How active are you?' (step 5 of 6).
Expected result	The bar shows about 83 % and reaches 100 % only on 'Here is your plan'.
Actual result	The bar is already 100 % on the activity step.
Impact	Cosmetic; the user is told they are finished one screen early.
Evidence	TC-014_activity_step
Reporter / Date	Usman Ghani · 11-Oct-2026

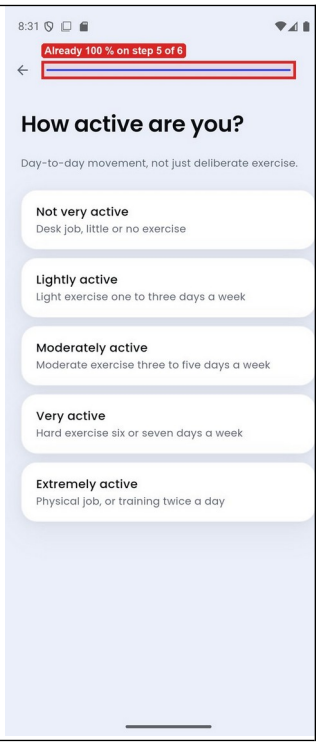


Figure 7: DEF-05 evidence: TC-014_activity_step (highlight added)

DEF-06 – Underweight users' offered alternatives end on an endless loading spinner

Defect ID	DEF-06
Title	Underweight users' offered alternatives end on an endless loading spinner
Module / Feature	M2 Goal Engine & Safety Rules
Found by	TC-018, TC-077
Severity / Priority	Critical / High
Status	Open
Steps to reproduce	1. Clear data; onboard as Male, 30, 178 cm, 52 kg. 2. On the target step enter 56 kg (refused). 3. Tap 'Aim for a healthy weight instead'; choose 'Moderately active'. 4. Repeat with 55 kg -> target 52 kg and 'Maintain instead'.
Expected result	A plan is shown: a gain toward about 59 kg, or a maintenance plan of about 2,350 kcal, with 'Start tracking'.
Actual result	The plan screen shows only a spinning loader - no target, no button, no back arrow. It never finishes (checked after 15 s); Back closes the app.
Impact	Any user under BMI 18.5 cannot finish onboarding at all, using the very options the app recommends.
Evidence	TC-018_spinner_after_15s, TC-018_refused_gain, TC-017_maintain_spinner
Reporter / Date	Usman Ghani · 11-Oct-2026

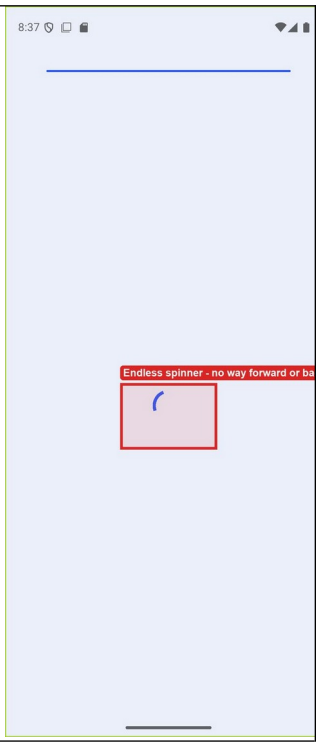


Figure 8: DEF-06 evidence: TC-018_spinner_after_15s (highlight added)

DEF-07 – 'Too fast' correction says 'losing' on a weight-gain goal

Defect ID	DEF-07
Title	'Too fast' correction says 'losing' on a weight-gain goal
Module / Feature	M2 Goal Engine & Safety Rules
Found by	TC-019
Severity / Priority	Minor / Medium
Status	Open
Steps to reproduce	1. Onboard Male, 30, 178 cm, 70 kg. 2. Enter target 85 kg with 12 weeks.
Expected result	The correction says gaining more than 1 % a week is too fast.
Actual result	'That timeframe would mean losing more than 1% of your bodyweight a week…' - for a gain goal. Shown on the target step and the plan.
Impact	Contradictory advice on the screen whose purpose is to explain the plan.
Evidence	TC-019_gain_too_fast, TC-019_plan
Reporter / Date	Usman Ghani · 11-Oct-2026

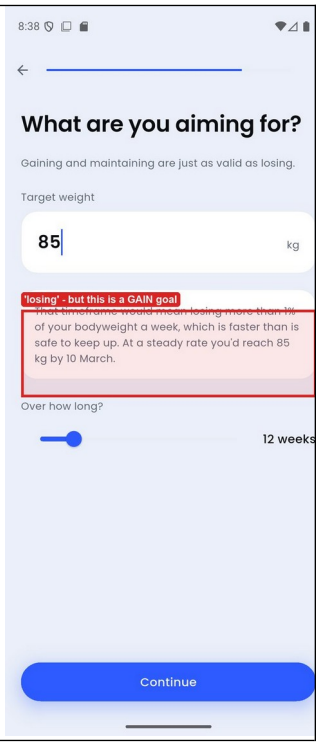


Figure 9: DEF-07 evidence: TC-019_gain_too_fast (highlight added)

DEF-11 – 'Add to log' stays stuck on 'Saving...' after the first manual add

Defect ID	DEF-11
Title	'Add to log' stays stuck on 'Saving...' after the first manual add
Module / Feature	M3 Manual Food Logging
Found by	TC-034
Severity / Priority	Major / High
Status	Open
Steps to reproduce	1. Tap '+', search 'rice', expand a result, tap 'Add to log' (returns to Home). 2. Tap '+' again and look at the same result card; tap its button.
Expected result	The card is back at rest and the food can be added again.
Actual result	The card is still expanded with a disabled 'Saving...' button. Tapping does nothing; it stays stuck for the session.
Impact	Repeat foods - the most common manual log - cannot be added again without a new search.
Evidence	TC-034_stuck_saving, V1_stuck_saving_button.mp4
Reporter / Date	Faizan Ali · 11-Oct-2026
Note	The project's next commit (36ead9d) says it fixes this; not re-tested in this cycle.

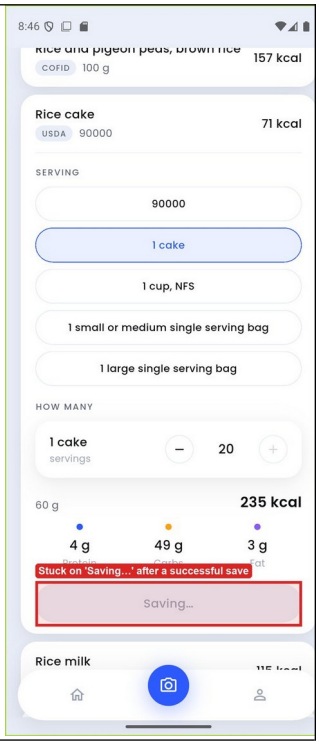


Figure 10: DEF-11 evidence: TC-034_stuck_saving (highlight added)

DEF-12 – Recent foods never appear for foods logged by hand

Defect ID	DEF-12
Title	Recent foods never appear for foods logged by hand
Module / Feature	M3 Manual Food Logging
Found by	TC-030
Severity / Priority	Minor / Medium
Status	Open
Steps to reproduce	1. Log a food through Add food. 2. Open Add food with an empty search field.
Expected result	A 'Recent' section lists recently/frequently logged foods (UC-05 step 2).
Actual result	Nothing is shown. Only foods saved from photo estimates ever appear as Recent (seen during photo-logging tests).
Impact	Manual-only users - the default build - never get the shortcut UC-05 promises.
Evidence	TC-030_no_recent_section
Reporter / Date	Faizan Ali · 11-Oct-2026

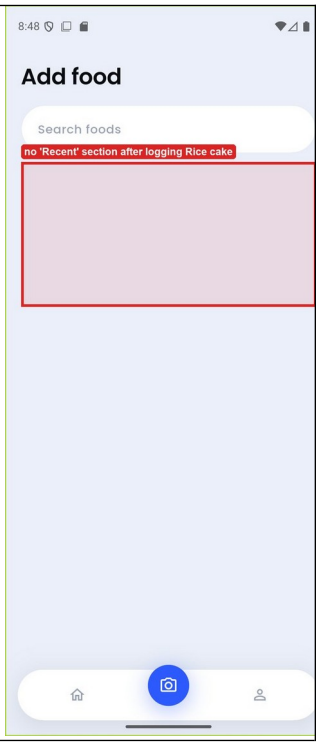


Figure 11: DEF-12 evidence: TC-030_no_recent_section (highlight added)

DEF-13 – Implausible nutrition values for some INDB foods

Defect ID	DEF-13
Title	Implausible nutrition values for some INDB foods
Module / Feature	M3 Manual Food Logging
Found by	EXP-02
Severity / Priority	Major / Medium
Status	Open
Steps to reproduce	1. Search 'palak channa cutlet', expand 'Spinach chickpeas cutlet (Palak channa dal cutlet)'. 2. Search 'egg boiled', look at 'Boiled egg (Ubla anda)'.
Expected result	One cutlet is roughly 60–100 g and 150–250 kcal; one boiled egg is about 50 g and 70–80 kcal.
Actual result	1 cutlet = 328 g, 2,255 kcal, 241 g fat (73 % fat by weight). 'Boiled egg (Ubla anda)' 1 egg = 151 g but 68 kcal (45 kcal/100 g).
Impact	One log of the cutlet adds a whole day's calories; totals and targets become meaningless.
Evidence	TC-EXP_cutlet_2255kcal, TC-039_editor_open
Reporter / Date	Faizan Ali · 11-Oct-2026

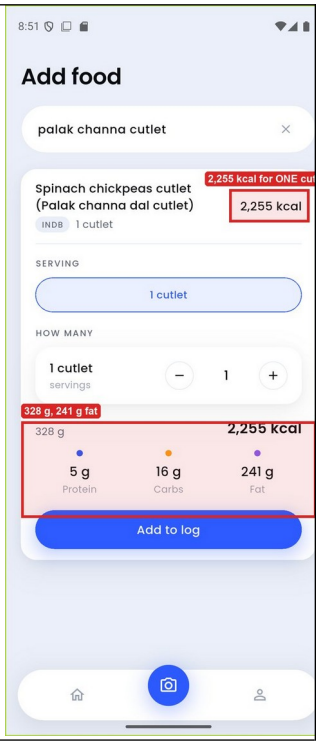


Figure 12: DEF-13 evidence: TC-EXP_cutlet_2255kcal (highlight added)

DEF-14 – Portion chip labelled '0.8×' applies 0.75×

Defect ID	DEF-14
Title	Portion chip labelled '0.8×' applies 0.75×
Module / Feature	M4 Day View & Entry Editing
Found by	EXP-03
Severity / Priority	Minor / Low
Status	Open
Steps to reproduce	1. Open an entry (e.g. 151 g boiled egg). 2. Tap '0.8×'.
Expected result	The item becomes 0.8 × 151 g = 121 g.
Actual result	It becomes 113 g / 51 kcal (0.75×). The label rounds 0.75 to '0.8'.
Impact	The amount logged differs from the amount chosen.
Evidence	EXP_chip_0_8x
Reporter / Date	Faizan Ali · 11-Oct-2026

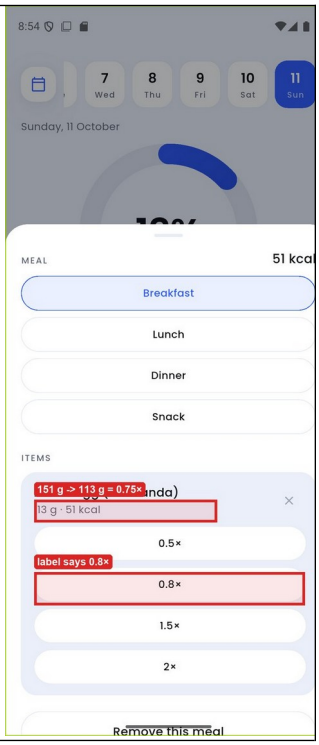


Figure 13: DEF-14 evidence: EXP_chip_0_8x (highlight added)

DEF-17 – Calendar day detail omits fat

Defect ID	DEF-17
Title	Calendar day detail omits fat
Module / Feature	M5 Calendar
Found by	TC-049
Severity / Priority	Trivial / Low
Status	Open
Steps to reproduce	1. Open Calendar, tap a logged day.
Expected result	Protein, carbs and fat are listed.
Actual result	Only Protein and Carbs are shown.
Impact	Incomplete day summary.
Evidence	TC-049_day_detail
Reporter / Date	Faizan Ali · 11-Oct-2026

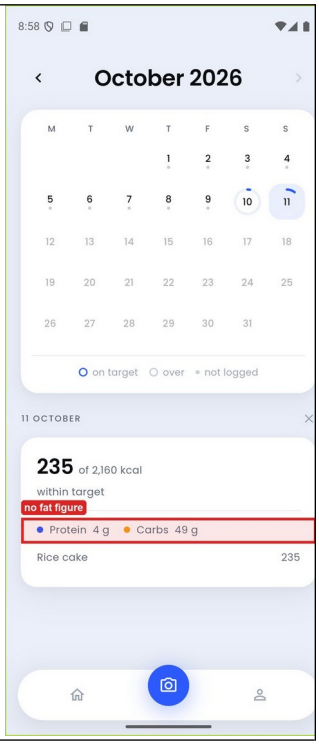


Figure 14: DEF-17 evidence: TC-049_day_detail (highlight added)

DEF-21 – Home ring % (and Calendar) keep the old target after a goal change

Defect ID	DEF-21
Title	Home ring % (and Calendar) keep the old target after a goal change
Module / Feature	M4 Day View & Entry Editing
Found by	TC-045
Severity / Priority	Major / High
Status	Open
Steps to reproduce	1. Log 235 kcal today (ring 11 % of 2,160). 2. Goal: set calories by hand to 1,080. 3. Return to Home.
Expected result	Ring 22 % and line '235 / 1,080 kcal'.
Actual result	The line updates to '235 / 1,080 kcal' but the ring still says 11 %. It corrects only after switching day; the Calendar keeps the old target until restart.
Impact	Two figures on the same screen disagree about progress.
Evidence	TC-045_home_after_goal_change, TC-045_home_before, TC-045_today_after_day_switch
Reporter / Date	Faizan Ali · 11-Oct-2026

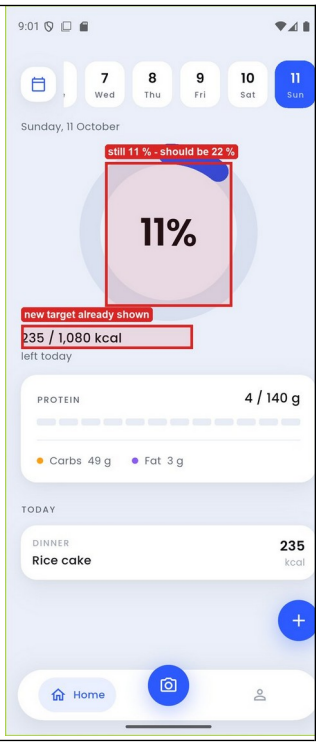


Figure 15: DEF-21 evidence: TC-045_home_after_goal_change (highlight added)

DEF-22 – Past days are re-judged against a new goal

Defect ID	DEF-22
Title	Past days are re-judged against a new goal
Module / Feature	M5 Calendar
Found by	TC-052
Severity / Priority	Major / High
Status	Open
Steps to reproduce	1. Log 72 kcal on 10 Oct against 2,160 kcal (3 %). 2. Goal: set calories to 1,080. 3. View 10 Oct on Home, and in Calendar after a restart.
Expected result	Past days keep the target in force when logged (UC-10; Goal screen: 'Days you have already logged keep the target they were measured against').
Actual result	10 Oct now reads '72 / 1,080 kcal' (7 %) on Home and '72 of 1,080 kcal' in Calendar.
Impact	History changes retroactively, breaking a stated promise and any record of adherence.
Evidence	TC-052_calendar_yesterday_after_restart, TC-052_yesterday_home_after_change, TC-052_calendar_yesterday
Reporter / Date	Faizan Ali · 11-Oct-2026

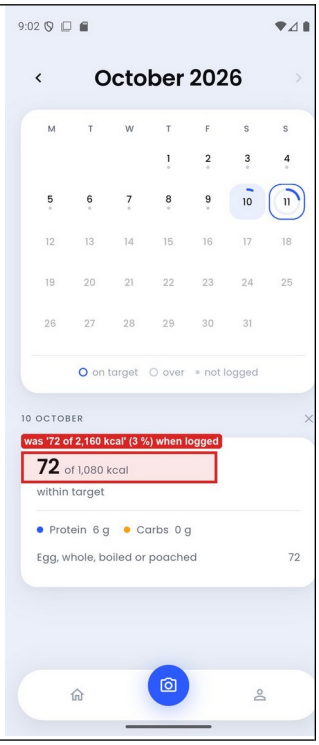


Figure 16: DEF-22 evidence: TC-052_calendar_yesterday_after_restart (highlight added)

DEF-23 – Switching the weight unit erases the goal date and hand-set calories

Defect ID	DEF-23
Title	Switching the weight unit erases the goal date and hand-set calories
Module / Feature	M6 Goal Management
Found by	TC-061
Severity / Priority	Major / High
Status	Open
Steps to reproduce	1. Goal: loss goal 'On track … by 3 January', rate 0.6 %; set calories by hand to 2,000. 2. Under About -> Weight tap 'lb'. 3. Force-stop and relaunch; open Goal.
Expected result	Only the display unit changes; caption, rate, target and the hand-set value stay.
Actual result	Caption becomes 'Enough to hold your current weight.' (goal still 172 -> 159 lb), rate drops to 0.5 %, and after restart the 'Calori suggests… Reset' line is gone - the saved target date and hand-set flag were wiped.
Impact	A display preference silently rewrites the saved plan.
Evidence	TC-061_after_restart, TC-061_before_units_toggle, TC-061_after_lb, V2_units_toggle_wipes_goal.mp4
Reporter / Date	Faizan Ali · 11-Oct-2026

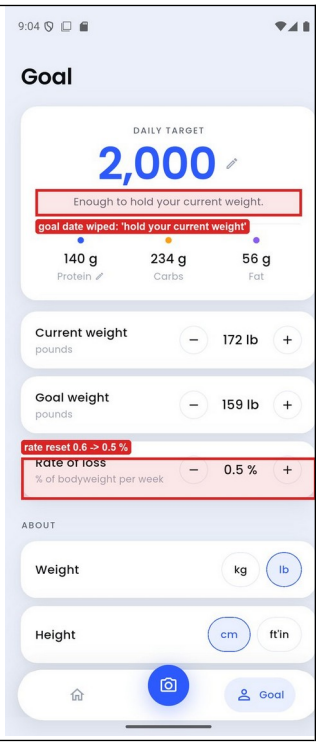


Figure 17: DEF-23 evidence: TC-061_after_restart (highlight added)

DEF-24 – Goal screen ignores the chosen rate (target up to 220 kcal off)

Defect ID	DEF-24
Title	Goal screen ignores the chosen rate (target up to 220 kcal off)
Module / Feature	M6 Goal Management
Found by	TC-056
Severity / Priority	Major / High
Status	Open
Steps to reproduce	1. Goal: current weight 95 kg, goal 92 kg (Male, 30, 178 cm, moderately active). 2. Set 'Rate of loss' to 1.0 %.
Expected result	Target 1,930 kcal; 'At 1.0 % per week you reach your target by 3 November'.
Actual result	'At 0.8 % per week you reach your target by 8 November', 2,150 kcal - 220 kcal more than the chosen rate implies.
Impact	The core calculation does not honour the user's input.
Evidence	TC-056_rate_1_0_quantised, TC-055_rate_max
Reporter / Date	Faizan Ali · 11-Oct-2026
Note	The project's next commit (36ead9d) says it fixes this; not re-tested in this cycle.

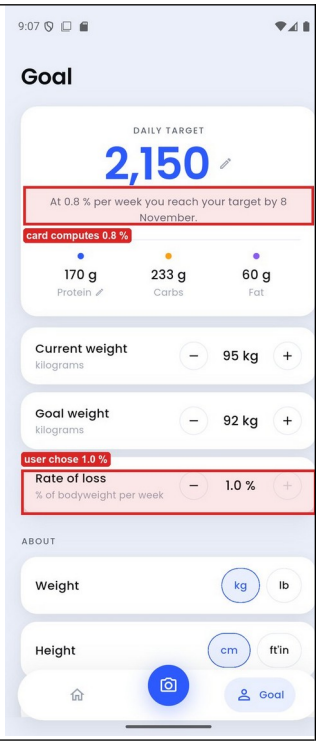


Figure 18: DEF-24 evidence: TC-056_rate_1_0_quantised (highlight added)

DEF-25 – Viewfinder's Gallery button has no visible label

Defect ID	DEF-25
Title	Viewfinder's Gallery button has no visible label
Module / Feature	M7 Photo Logging
Found by	TC-063
Severity / Priority	Major / Medium
Status	Open
Steps to reproduce	1. Tap the camera button to open the viewfinder. 2. Look for the Gallery button beside the shutter.
Expected result	A labelled 'Gallery' pill beside the shutter.
Actual result	The Gallery pill stretches across the whole bar and its 'Gallery' text is hidden behind the shutter; the bar looks empty.
Impact	Choosing an existing photo is effectively undiscoverable.
Evidence	TC-063_viewfinder
Reporter / Date	Usman Ghani · 11-Oct-2026



Figure 19: DEF-25 evidence: TC-063_viewfinder (highlight added)

DEF-26 – Viewfinder's Close (×) sits under the status bar and cannot be tapped

Defect ID	DEF-26
Title	Viewfinder's Close (×) sits under the status bar and cannot be tapped
Module / Feature	M7 Photo Logging
Found by	TC-063
Severity / Priority	Major / High
Status	Open
Steps to reproduce	1. Open the viewfinder. 2. Tap the × in the top-left corner.
Expected result	The viewfinder closes.
Actual result	Nothing happens: the button (y 32–147 px) lies under the status bar (y 0–142 px). Only a tap in the 5 px strip at y = 145 worked.
Impact	Users are trapped in the camera unless they know to use system Back.
Evidence	TC-063_viewfinder, TC-063_after_close
Reporter / Date	Usman Ghani · 11-Oct-2026



Figure 20: DEF-26 evidence: TC-063_viewfinder (highlight added)

DEF-27 – Proposed photo items' portions cannot be adjusted before saving

Defect ID	DEF-27
Title	Proposed photo items' portions cannot be adjusted before saving
Module / Feature	M7 Photo Logging
Found by	TC-066
Severity / Priority	Major / High
Status	Open
Steps to reproduce	1. Analyse biryani.jpg from the Gallery. 2. Try to change any item's portion (tap '250 g · 5 pieces').
Expected result	Every proposed item's portion can be adjusted, with live totals (UC-06 main flow).
Actual result	No control exists; portion chips appear only for low-confidence items and none were offered here. Only × (remove) is available.
Impact	A wrong estimate must be saved first and fixed later, or discarded.
Evidence	TC-066_portion_tap_no_control
Reporter / Date	Usman Ghani · 11-Oct-2026

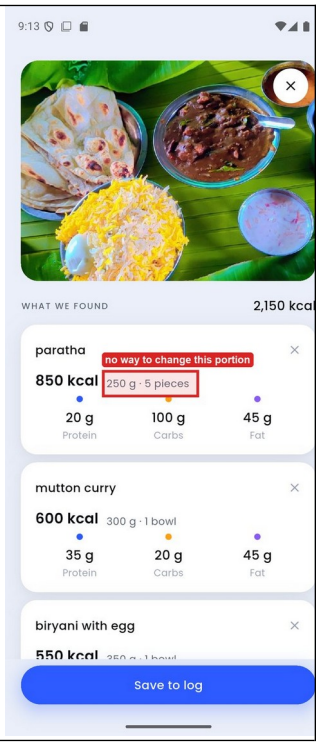


Figure 21: DEF-27 evidence: TC-066_portion_tap_no_control (highlight added)

DEF-29 – After saving a gallery photo the app returns to the camera, not Home

Defect ID	DEF-29
Title	After saving a gallery photo the app returns to the camera, not Home
Module / Feature	M7 Photo Logging
Found by	TC-064
Severity / Priority	Minor / Medium
Status	Open
Steps to reproduce	1. Viewfinder -> Gallery -> choose a meal photo. 2. Tap 'Save to log'.
Expected result	'Meal added' and Home, with the totals updated (UC-03).
Actual result	'Meal added', then the live viewfinder again. Combined with DEF-26 the user must use system Back to leave.
Impact	Confusing flow; invites logging the meal twice.
Evidence	TC-064_meal_added_snackbar, TC-064_home_after_save
Reporter / Date	Usman Ghani · 11-Oct-2026

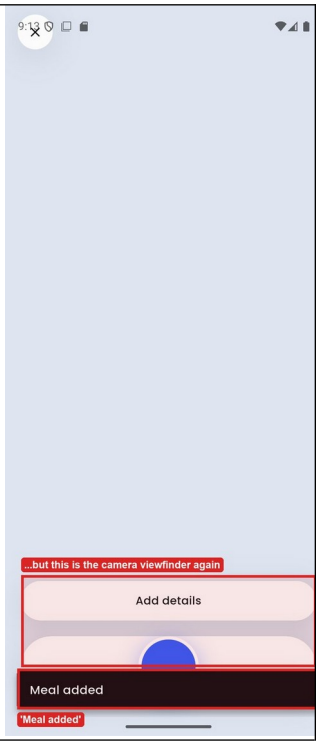


Figure 22: DEF-29 evidence: TC-064_meal_added_snackbar (highlight added)

DEF-30 – Details typed before choosing a gallery photo are ignored

Defect ID	DEF-30
Title	Details typed before choosing a gallery photo are ignored
Module / Feature	M7 Photo Logging
Found by	TC-065
Severity / Priority	Major / Medium
Status	Open
Steps to reproduce	1. Viewfinder -> Add details 'three apples, about 540 g' -> Done. 2. Gallery -> apple.jpg (one apple).
Expected result	The estimate follows the user's details (about 3 apples / 540 g) (UC-04).
Actual result	'apple · 95 kcal · 182 g · 1 medium apple' - the details were silently dropped.
Impact	The accuracy feature UC-04 exists for does nothing on the gallery path.
Evidence	TC-065_apple_proposal, TC-065_details_set
Reporter / Date	Usman Ghani · 11-Oct-2026

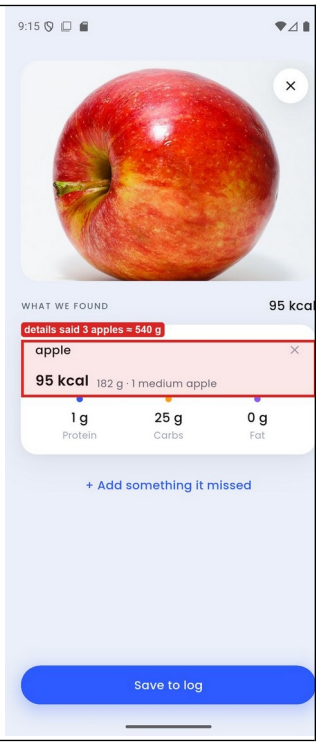


Figure 23: DEF-30 evidence: TC-065_apple_proposal (highlight added)

DEF-33 – Fields meant to take focus do not (Add food search, target weight)

Defect ID	DEF-33
Title	Fields meant to take focus do not (Add food search, target weight)
Module / Feature	M3 Manual Food Logging
Found by	EXP-01
Severity / Priority	Trivial / Low
Status	Open
Steps to reproduce	1. Tap '+' on Home. 2. Or reach 'What are you aiming for?' in onboarding.
Expected result	The field is focused with the keyboard up, ready to type.
Actual result	No focus and no keyboard; the user must tap the field first.
Impact	One extra tap on the most frequent action.
Evidence	TC-075_plus_add_food
Reporter / Date	Faizan Ali · 11-Oct-2026



Figure 24: DEF-33 evidence: TC-075_plus_add_food (highlight added)

DEF-34 – Past days say 'left today'

Defect ID	DEF-34
Title	Past days say 'left today'
Module / Feature	M4 Day View & Entry Editing
Found by	EXP-06
Severity / Priority	Trivial / Low
Status	Open
Steps to reproduce	1. On Home select an earlier day that has entries (e.g. 10 Oct).
Expected result	Wording fits a past day (e.g. 'left' or 'under target').
Actual result	'72 / 2,160 kcal' followed by 'left today' on Saturday, 10 October.
Impact	Cosmetic wording error.
Evidence	TC-035_logged_on_10oct
Reporter / Date	Faizan Ali · 11-Oct-2026

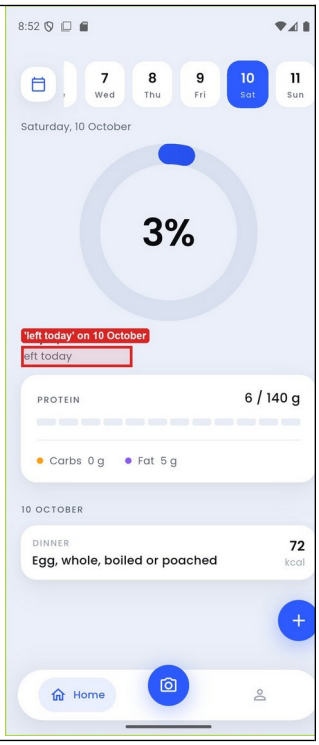


Figure 25: DEF-34 evidence: TC-035_logged_on_10oct (highlight added)

Appendix E – Exploratory Testing Notes

Six time-boxed sessions were run alongside the scripted cases, each with a charter. Findings that are functional defects are logged in Appendix D; usability observations are recorded here only.

Session	EXP-01
Charter	Onboarding: navigation, persistence and focus outside the scripted paths
Tester	Usman Ghani
Time (11-Oct-2026)	20:14–20:45
Findings	System Back on any onboarding step closes the app and loses all answers (DEF-03). Restarting after the under-18 refusal returns to step 1 (DEF-02). Progress bar is full on the activity step, one step early (DEF-05). Target-weight field does not take focus on arrival (DEF-33). Observation (usability, out of scope): deleting digits passes through refused values, so the keyboard auto-closes 0.9 s after an intermediate refusal while editing.

Session	EXP-02
Charter	Food database: data quality while browsing search results
Tester	Faizan Ali
Time (11-Oct-2026)	20:45–20:52
Findings	INDB 'Spinach chickpeas cutlet' 1 cutlet = 328 g / 2,255 kcal / 241 g fat; 'Boiled egg (Ubla anda)' 151 g = 68 kcal (DEF-13). Observation: some CNF rice dishes use a liquid serving ('100ml') for a solid food.

Session	EXP-03
Charter	Entry editor: every multiplier chip and repeated use
Tester	Faizan Ali
Time (11-Oct-2026)	20:52–20:55
Findings	'0.8×' applies 0.75× (DEF-14).

Session	EXP-04
Charter	Goal screen and shell: layout of controls near the floating nav bar
Tester	Faizan Ali
Time (11-Oct-2026)	21:05–21:09
Findings	Before scrolling, 'Where the food data comes from' sits under the floating nav bar; tapping it hit the camera button and raised the camera permission prompt. The round '+' button covers the kcal figure of the last meal card on Home. (Both are layout/usability observations, recorded but not counted as functional defects.)

Session	EXP-05
Charter	Viewfinder and photo flow: controls, exits and return paths
Tester	Usman Ghani
Time (11-Oct-2026)	21:08–21:21
Findings	Gallery label hidden under the shutter (DEF-25). Close (×) untappable under the status bar (DEF-26). After saving a gallery photo the app returns to the camera (DEF-29).

Session	EXP-06
Charter	Day view on past dates
Tester	Faizan Ali
Time (11-Oct-2026)	20:52–20:53
Findings	Past days read 'left today' (DEF-34).

Appendix F – Evidence Index

All screenshots are full-resolution PNG files (1080 × 2424) in the 'evidence' folder submitted with this report; screen videos are in the 'videos' folder.

F.1 Screen videos

Table 27: Screen videos

File	Shows	Linked to	Length
V1_stuck_saving_button.mp4	'Saving…' button stays disabled; tapping it does nothing; Home has one entry	TC-034, DEF-11	20 s
V2_units_toggle_wipes_goal.mp4	Tapping 'lb' changes the caption to 'hold your current weight' and the rate to 0.5 %	TC-061, DEF-23	9 s

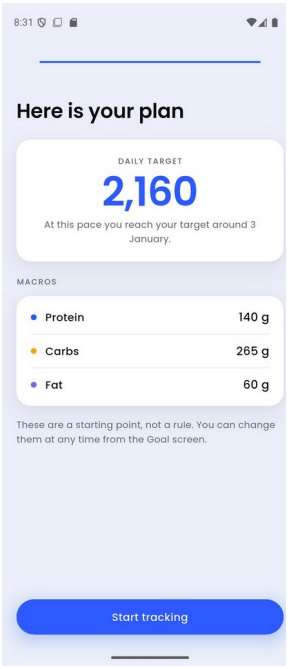
F.2 Screenshot index

Table 28: Screenshot index

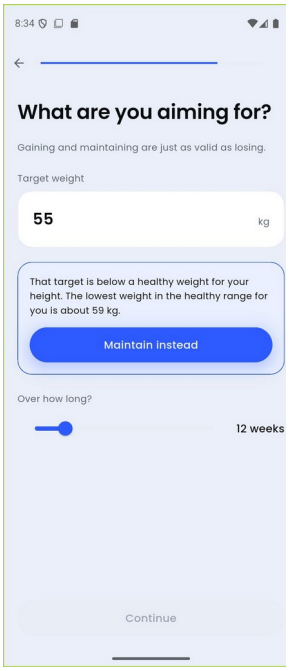
File	Linked to	Captured
EXP_camera_permission_prompt.png	TC-063	12:27
EXP_chip_0_8x.png	DEF-14	12:27
EXP_stale_results_empty_field.png	exploratory / working shot	12:27
TC-002_age_step.png	TC-002	12:27
TC-003_age18_body_step.png	TC-003	12:27
TC-003_age_119.png	TC-003	12:27
TC-004_age17_blocked.png	TC-004	12:27
TC-006_target_prefilled.png	TC-006	12:27
TC-010_back_to_cm.png	TC-010	12:27
TC-010_ftin_converted.png	TC-010	12:27
TC-010_lb_converted.png	TC-010	12:27
TC-012_before_back.png	DEF-03	12:27
TC-012_system_back_onboarding.png	DEF-03	12:27
TC-013_after_restart.png	DEF-02, TC-013	12:27
TC-013_reentered_age25.png	DEF-02, TC-013	12:27
TC-014_activity_step.png	DEF-05, TC-014	12:27
TC-014_plan.png	TC-014	12:27
TC-014_target_note.png	TC-014	12:27
TC-015_plan.png	TC-015	12:27
TC-015_rate_clamped.png	TC-015	12:27
TC-016_maintain_plan.png	TC-016	12:27
TC-016_refused_55.png	TC-016	12:27
TC-017_maintain_spinner.png	DEF-06, TC-077	12:27
TC-018_plan_after_healthy_weight.png	TC-018	12:27
TC-018_refused_gain.png	DEF-06, TC-018	12:27
TC-018_spinner_after_15s.png	DEF-06, TC-018	12:27
TC-019_gain_too_fast.png	DEF-07, TC-019	12:27
TC-019_plan.png	DEF-07, TC-019	12:27
TC-020_floor_plan.png	TC-020	12:27
TC-021_tdee_below_floor_plan.png	TC-021	12:27
TC-022_78_5_maintain.png	TC-022	12:27
TC-022_78_6_gain.png	TC-022	12:27
TC-025_home_after_relaunch.png	TC-025	12:27
TC-025_home_after_start.png	TC-025	12:27
TC-027_channa.png	TC-027	12:27
TC-027_karela.png	TC-027	12:27
TC-028_cleared.png	TC-028	12:27
TC-028_no_matches.png	TC-028	12:27
TC-030_no_recent_section.png	DEF-12, TC-030	12:27
TC-032_one_cake_qty2.png	TC-032	12:27
TC-032_qty2.png	TC-032	12:27
TC-033_home_after_add.png	TC-033	12:27
TC-033_snackbar.png	TC-033	12:27
TC-034_stuck_saving.png	DEF-11, TC-034	12:27
TC-035_logged_on_10oct.png	DEF-34	12:27
TC-036_home_two_entries.png	TC-036	12:27
TC-037_over_target.png	TC-037	12:27
TC-038_strip_newest.png	TC-038	12:27
TC-038_strip_oldest.png	TC-038	12:27
TC-039_breakfast_selected.png	TC-039	12:27

File	Linked to	Captured
TC-039_editor_open.png	DEF-13, TC-039	12:27
TC-039_home_breakfast_label.png	TC-039	12:27
TC-044_after_remove.png	TC-044	12:27
TC-044_editor.png	TC-044	12:27
TC-045_home_after_goal_change.png	DEF-21, TC-045	12:27
TC-045_home_before.png	DEF-21, TC-045	12:27
TC-045_today_after_day_switch.png	DEF-21, TC-045	12:27
TC-046_calendar_october.png	TC-046, TC-051	12:27
TC-048_september.png	TC-048	12:27
TC-049_closed.png	TC-049	12:27
TC-049_day_detail.png	DEF-17, TC-049	12:27
TC-052_calendar_yesterday.png	DEF-22, TC-052	12:27
TC-052_calendar_yesterday_after_restart.png	DEF-22, TC-052	12:27
TC-052_yesterday_home_after_change.png	DEF-22, TC-052	12:27
TC-053_goal_screen.png	TC-053	12:27
TC-055_rate_max.png	DEF-24	12:27
TC-056_rate_1_0_quantised.png	DEF-24, TC-056	12:27
TC-057_after_restart.png	TC-057	12:27
TC-057_refused_58.png	TC-057	12:27
TC-058_after_reset.png	TC-058	12:27
TC-058_kcal_1499_advisory.png	TC-058	12:27
TC-058_kcal_1500.png	TC-058	12:27
TC-058_kcal_sheet.png	TC-058	12:27
TC-061_after_lb.png	DEF-23, TC-061	12:27
TC-061_after_restart.png	DEF-23, TC-061	12:27
TC-061_before_units_toggle.png	DEF-23, TC-061	12:27
TC-062_sources.png	TC-062	12:27
TC-062_sources_2.png	TC-062	12:27
TC-063_after_close.png	DEF-26, TC-063	12:27
TC-063_viewfinder.png	DEF-25, DEF-26, TC-063	12:27
TC-064_analysing.png	exploratory / working shot	12:27
TC-064_entry_with_photo.png	TC-064	12:27
TC-064_home_after_save.png	DEF-29, TC-064	12:27
TC-064_meal_added_snackbar.png	DEF-29, TC-064	12:27
TC-064_picker.png	TC-064	12:27
TC-064_proposed.png	TC-064	12:27
TC-064_proposed_2.png	TC-064	12:27
TC-065_apple_proposal.png	DEF-30, TC-065	12:27
TC-065_details_set.png	DEF-30, TC-065	12:27
TC-065_details_typed.png	TC-065	12:27
TC-066_portion_tap_no_control.png	DEF-27, TC-066	12:27
TC-069_cat_no_food.png	TC-069	12:27
TC-070_offline_message.png	TC-070	12:27
TC-070_retry_result.png	TC-070	12:27
TC-072_biryani_repeat.png	TC-072	12:27
TC-074_goal_tab.png	TC-074	12:27
TC-074_home_tab.png	TC-074	12:27
TC-075_plus_add_food.png	DEF-33	12:27
TC-076_back_from_calendar.png	DEF-03, TC-076	12:27
TC-EXP_cutlet_2255kcal.png	DEF-13	12:27

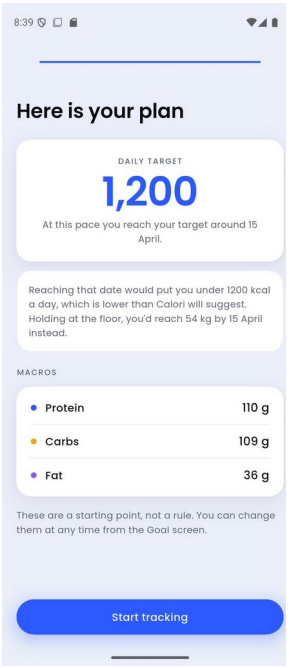
F.3 Key passed tests



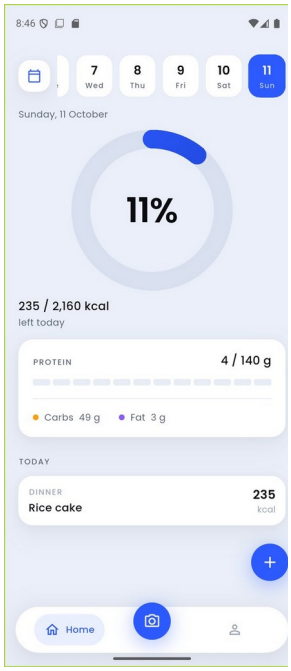
TC-014 reference plan



TC-016 BMI refusal

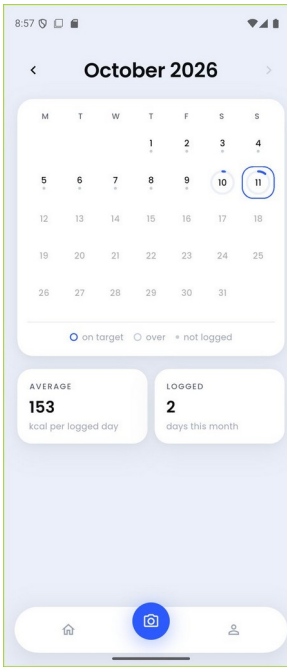


TC-020 calorie floor

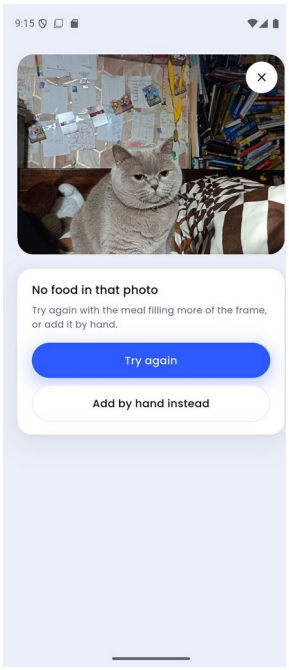


TC-033 food added

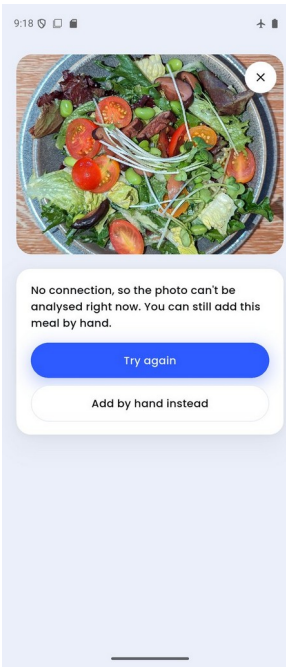
Figure 26: Key passed tests – onboarding and logging



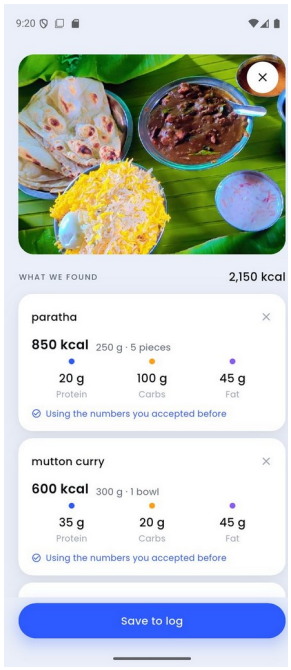
TC-046 calendar



TC-069 no food



TC-070 offline



TC-072 repeat meal

Figure 27: Key passed tests – calendar and photo path

Appendix G – Contribution Table

Table 29: Contribution by member

Member	Roll no.	Modules	TCs designed & executed	Defects reported	Exploratory sessions	Other work
Usman Ghani	24f3000	M1, M2, M7	23	9	2	Plan, environment, report compilation
Faizan Ali	24f3029	M3, M4, M5, M6, M8	25	12	4	RTM, defect log, report review

Appendix H – Reflection

Testing a real application against its own written use cases taught us more than testing a toy example could. Three lessons stand out.

Expected results must come from somewhere other than the app. Writing a small calculator from the published formula before running anything meant that a wrong target could not look right to us. That is how the Goal screen's rate problem (DEF-24) showed up as a clear 220 kcal gap rather than a number we accepted.

The edges are where the defects are. Most failures came from boundaries and transitions rather than the happy path: an underweight user taking the offered alternative, restarting after a refusal, pressing Android Back, adding the same food twice, or toggling a unit with a hand-set target. Decision tables and state transition tables made us list these paths deliberately.

Evidence has to be planned. A screenshot overwritten by the next step cost us a re-check; naming every capture after its test case, and highlighting the fault on the image, made the defect reports far easier to write and to read.

If we repeated the exercise we would test on a physical phone as well as the emulator, budget time for re-testing fixed builds, and automate the goal-engine checks so they could be re-run on every change.